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Identification of Intermediate-mass Black Hole Candidates among a Sample of Sd Galaxies

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Abstract

We analyzed images of every northern hemisphere Sd galaxy listed in the Third Reference Catalogue of Bright Galaxies with a relatively face-on inclination ($\theta \leqslant 30°$). Specifically, we measured the spiral arms' winding angle, ϕ, in 85 galaxies. We applied a novel black hole mass planar scaling relation involving the rotational velocities (from the literature) and pitch angles of each galaxy to predict central black hole masses. This yielded 23 galaxies, each having at least a 50% chance of hosting a central intermediate-mass black hole (IMBH), $10^2 < M_\bullet \leqslant 10^5\,M_\odot$. These 23 nearby ($\lesssim$50 Mpc) targets may be suitable for an array of follow-up observations to check for active nuclei. Based on our full sample of 85 Sd galaxies, we estimate that the typical Sd galaxy (which tends to be bulgeless) harbors a black hole with $\log(M_\bullet/M_\odot) = 6.00 \pm 0.14$, but with a 27.7% chance of hosting an IMBH, making this morphological type of galaxy fertile ground for hunting elusive IMBHs. Thus, we find that a $\sim 10^6\,M_\odot$ black hole corresponds roughly to the onset of bulge development and serves as a conspicuous waypoint along the galaxy–supermassive black hole coevolution journey. Our survey suggests that >1.22% of bright galaxies ($B_T \lesssim 15.5$ mag) in the local Universe host an IMBH (i.e., the "occupation fraction"), which implies a number density $>4.96 \times 10^{-6}$ Mpc^{-3} for central IMBHs. Finally, we observe that Sd galaxies exhibit an unexpected diversity of properties that resemble the general population of spiral galaxies, albeit with an enhanced signature of the eponymous prototypical traits (i.e., low masses, loosely wound spiral arms, and smaller rotational velocities).

Unified Astronomy Thesaurus concepts: Astrostatistics (1882); Galaxy evolution (594); Hubble classification scheme (757); Intermediate-mass black holes (816); Late-type galaxies (907); Regression (1914); Scaling relations (2031); Spiral galaxies (1560); Spiral pitch angle (1561)

Materials only available in the online version of record: animation

1. Introduction

Intermediate-mass black holes (IMBHs) could be characterized as the rare larval form of black holes. Announced within a span of just 3 yr, astronomers built upon a century of black hole research[9] to obtain emphatic evidence for the embryonic and adult forms of black holes: stellar mass (Abbott et al. 2016) and supermassive (Event Horizon Telescope Collaboration et al. 2019, 2022) black holes (SMBHs), respectively. However, speculation abounds concerning the intervening IMBH category that bridges the mass gap between stellar mass and SMBHs. The gap in our knowledge is akin to a boxing league that has lightweight and heavyweight divisions, but no middleweight division, in which there are three possibilities: (i) individual boxers have always been either lightweight or heavyweight, (ii) middleweight boxers always gain weight rapidly to become heavyweights, or (iii) there is an underground fighting league for middleweights that we are not privy to. Thus, we are on a quest to find evidence of scenario (iii) by tracking down clandestine IMBHs in their galactic lairs.

Perhaps the first evidence of the missing demographical category of black holes came at the turn of the millennium with the identification of an IMBH with a mass $>700\,M_\odot$ in M82 (Matsushita et al. 2000; Ebisuzaki et al. 2001; Kaaret et al. 2001; Matsumoto et al. 2001). Still, almost a quarter-century later, obtaining direct confirmation of IMBHs is not easy (see Koliopanos 2017; Mezcua 2017; Greene et al. 2020, for recent reviews). Even if one could prognosticate with certainty where to look for an IMBH, the telescope resources and time investment to garner definitive proof of IMBHs are significant (see Graham et al. 2021b, their Section 5). It is for these reasons that reconnaissance work is required to preselect the galaxies that are likely to harbor IMBHs. Such scouting studies will become invaluable to the world's best observatories as they look for IMBHs.

Large surveys will surely play a pivotal role in the search for IMBHs. Data mining of preexisting surveys can be a promising avenue for the selection of candidates. Furthermore, large survey telescopes like the Vera C. Rubin Observatory (née Large Synoptic Survey Telescope; Tyson 2002) and the

[9] See Graham (2016a) for key results and a detailed timeline of black hole research.

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Einstein Probe (Yuan et al. 2022) are expected to find vast numbers of active galactic nuclei (AGNs), tidal disruption events (TDEs), and likely IMBHs. Careful filtering will be required to sort through the immense quantity of data produced to efficiently identify IMBH candidates.

One of the best methods for predicting black hole masses in galaxies is via black hole mass scaling relations (e.g., Bennert et al. 2011; Graham et al. 2015; Graham 2016a; D'Onofrio et al. 2021; Izquierdo-Villalba et al. 2023). By identifying galaxies that reside at the same extreme end of multiple scaling relations, the success of their combined predictive power becomes more probable (Koliopanos et al. 2017; Graham & Soria 2019; Graham et al. 2019, 2021a, 2021b; Davis & Graham 2021). Furthermore, because extrapolating is inherently uncertain, it is safer to extrapolate from multiple relations and look for agreement rather than trusting that a single relation holds beyond its defined range. Therefore, we seek a class of galaxies that consistently occupies the extremities of separate black hole mass scaling relations. Moreover, we effectively entwine the separate scaling relations by applying a novel trivariate relationship (Davis & Jin 2023).

Late-type spiral (e.g., Sd) galaxies are the most compatible morphological class of galaxies for the focus of our experiment. Sd galaxies tick all the boxes in a census of galaxies that place them in a rare demographic category, which is highly likely to cohabit with IMBHs. We note that dwarf early-type galaxies (particularly low-mass S0 galaxies) are also good candidates to host IMBHs (Graham & Soria 2019). Furthermore, they have been shown to occasionally possess faint disk substructure, including bars and spiral arms, the latter of which can be quantified via pitch angle measurements (Jerjen et al. 2000; Lisker et al. 2006; Michea et al. 2021). However, the oft-hidden volute structure (*if* it exists) requires significant image processing to extract the embedded disk component, which itself has been historically missed even in bright early-type galaxies. Specifically, Lisker et al. (2006) found that 41 out of the 476 (8.6%) dwarf early-type galaxies in their sample showed "possible, probable, or unambiguous disk features." Although, the semblance of spiral structure in dwarf early-type galaxies could be caused by tidal triggering resulting from the cluster harassment of passive dwarf galaxies (Smith et al. 2021b).

Sd galaxies have open spiral structures, low total masses, small central velocity dispersions, slow rotational velocities, and are likely bulgeless; all such traits establish environments that should be ripe for the existence of IMBHs. Moreover, their structure is regular enough to not place them in peculiar or Magellanic-type distinctions. The preponderance of low-mass, disk-dominated, bulgeless galaxies in our sample also distinguishes our galaxies as having lived relatively merger-free lives. This creates an ideal scenario to conduct a clean test of IMBH growth in the absence of external influences on nuclear black holes that must have evolved in relative isolation (unlike Sm galaxies). Indeed, it is expected that the galaxy merger fraction monotonically increases as a function of stellar mass (Guzmán-Ortega et al. 2023) and bulge mass (Graham 2023a). Nonetheless, evidence shows that nonmerger processes alone are sufficient to fuel massive black hole growth in galaxies (Smethurst et al. 2021) and sustain coevolution between SMBHs and their host galaxies (Smethurst et al. 2024).

The ongoing pursuit of IMBHs has seen significant recent contributions. Using the Chandra X-ray Observatory (CXO; Weisskopf et al. 2000), the "Chandra Virgo Cluster Survey of Spiral Galaxies" (Soria et al. 2022, see also Chilingarian et al. 2018 and Bi et al. 2020) obtained long exposures for spiral galaxies in the Virgo Cluster. When combined with archival data, this project has X-ray imaging for all spiral galaxies in the Virgo Cluster with star formation rates $\gtrsim 1\,M_\odot\,\mathrm{yr}^{-1}$. Early identification of nuclear X-ray point sources from the archival CXO data, coupled with black hole mass scaling relations, yielded $3+11$ strong[10] IMBH candidates (Graham et al. 2019, 2021b). Karachentsev & Karachentseva (2019) collated a catalog of 220 face-on bulgeless galaxies, approximately half of which exhibit unresolved nuclei. Because nuclear star clusters (NSCs) are known to scale with their host galaxies (Balcells et al. 2003; Graham 2003; Wehner & Harris 2006; den Brok et al. 2014; Georgiev et al. 2016; Sánchez-Janssen et al. 2019; Pechetti et al. 2020) and their central black holes (Graham & Spitler 2009; Scott & Graham 2013; Graham 2016b; Georgiev et al. 2016; Graham 2020; Neumayer et al. 2020), such a catalog is a valuable reference of potential IMBH host galaxies.

Sd galaxies are certainly not the only place where IMBHs might exist. McKernan et al. (2012) envision an efficient process by which IMBHs may be efficiently grown in AGNs in the disks surrounding SMBHs. There might be IMBHs roaming our Galaxy (Schödel et al. 2005; Oka et al. 2017; Tsuboi et al. 2017; Takekawa et al. 2019; Tsuboi et al. 2019; Gravity Collaboration et al. 2020; Reid & Brunthaler 2020; Takekawa et al. 2020; Tsuboi et al. 2020; Zhu et al. 2020; Weller et al. 2022) or in its satellites that reveal themselves more readily than IMBHs in extragalactic environments.[11] Indeed, Nguyen et al. (2019) estimated a black hole mass ($M_\bullet \equiv M_{\rm BH}$) of $\log(M_\bullet/M_\odot) = 3.83^{+0.43}_{-0.60}$ in Messier 110 (a dwarf elliptical satellite of the Andromeda galaxy) via stellar dynamical modeling. Verily, not all fish are in the oceans but also rivers, ponds, and lakes. Analogously, Sd galaxies represent the major fishing areas for catching potential IMBHs. It is our endeavor here to establish, in general, (i) how common IMBHs might be in Sd galaxies and explicitly, (ii) which Sd galaxies pose the most promising potential for future studies to search for IMBHs.

We define our sample of Sd galaxies in Section 2, and apply a planar black hole mass scaling relation to the collection of galaxies (in Section 3). In Section 4, we present the results and statistics for our sample and identify our targets of interest. Additionally, we discuss other observational considerations such as AGNs and X-ray point sources. Finally, we provide a discussion (in Section 5) on our findings for Sd galaxies, detail the implications from IMBH research, consider the prospects of detecting IMBHs in our sample, consider the outlook for follow-up investigations, and summarize our findings in Section 6. We represent masses (M) throughout this work as logarithmic (solar) masses ($\mathcal{M}$), such that $\mathcal{M} \equiv \log(M/M_\odot)$. Stellar masses have all been adjusted to conform with the Chabrier (2003) initial mass function[12] and we assume cosmographic parameters from Planck Collaboration et al. (2020). All uncertainties are quoted at $1\sigma \equiv 68.3\%$ confidence intervals; median absolute deviations are given as uncertainties associated with medians.

[10] Here, *strong* candidates are those with both a predicted black hole mass $\lesssim 10^5\,M_\odot$ and a centrally located X-ray point source.

[11] Specifically, Paynter et al. (2021) estimated there are $\approx 4.6 \times 10^4$ IMBHs with masses between $\approx 10^4$ and $10^5\,M_\odot$ in the neighborhood of the Milky Way.

[12] See Davis et al. (2018, 2019b) for further details on how we homogenize stellar masses from various studies.

2. Sample of Galaxies

We assembled a sample of Sd (i.e., SAd, SABd, and SBd) type galaxies (Shapley & Paraskevopoulos 1940; de Vaucouleurs 1959; Graham 2019). We focused on the Sd, rather than Sc and earlier types, because of the expected larger spiral-arm winding angles; open spiral arms correlate with low black hole masses (Seigar et al. 2008; Berrier et al. 2013; Davis et al. 2017). The Magellanic-like Sm and Sdm (nor irregular) galaxies were not included because of their inherent disrupted and asymmetric structures.[13] Sd galaxies exhibit open spiral-arm structures and possess very faint (or no) bulges; they are ideal galaxies to host IMBHs. Because the bulge-to-total flux ratio (B/T) of Sd galaxies is low enough[14] to be considered "bulgeless,"[15] traditional (black hole mass)–(bulge mass) and (black hole mass)–(Sérsic index) relations are problematic, indicating a population of galaxies that likely possess small (or no) nuclear black holes.[16]

Bulgeless galaxies may still be analyzed with other black hole mass scaling relations that are not dependent on bulge masses, but some other physical property, such as spiral-arm pitch angle (Seigar et al. 2008; Berrier et al. 2013; Davis 2015; Davis et al. 2017), central velocity dispersion (Ferrarese & Merritt 2000; Gebhardt et al. 2000; Sahu et al. 2019a), total stellar mass (Beifiori et al. 2012; Davis et al. 2018; Sahu et al. 2019b; Graham 2023a),[17] and rotational velocity (Ferrarese 2002; Sabra et al. 2015; Davis et al. 2019c; Smith et al. 2021a). The total number of globular clusters bound to a galaxy is also an intriguing proxy for its central black hole mass (Burkert & Tremaine 2010; Harris & Harris 2011; Harris et al. 2014; González-Lópezlira et al. 2022). Notably, Bluck et al. (2023) find that the stellar gravitational potential of a galaxy is strongly correlated with SMBH mass, which is easy to measure in both observational and simulated data. The measurement of spiral-arm pitch angle (ϕ) requires a spiral galaxy first to be corrected to a face-on orientation via a deprojection of its apparent inclination with respect to the plane of the sky (see Davis et al. 2012, Figure 2). Because of this necessity, face-on galaxies are far easier to measure, with pitch angles becoming progressively harder to measure for more inclined galaxies and impossible for edge-on galaxies. For this reason, we have selected our sample of spiral galaxies to be close to face-on. Additionally, dust/inclination effects are minimized for galaxies when they are near to face-on.

We have constructed this study to take advantage of the remarkably low level of scatter for black hole mass estimates from the $M_\bullet$–ϕ relation. The usual go-to black hole mass scaling relation for the majority of studies has been the $M_\bullet$–σ_0 relation, due to its low level of scatter and availability of central velocity dispersion (σ_0) measurements. Indeed, central velocity dispersion is often described as the most fundamental black hole mass scaling relation parameter (e.g., de Nicola et al. 2019). However, this is predominantly the case for early-type galaxies; for late-type galaxies, the scatter is higher and σ_0 measurements are less common. From a common sample of 44 spiral galaxies, Davis et al. (2017) showed that the intrinsic scatter for the $M_\bullet$–σ_0 relation is 0.21 dex higher than that of the $M_\bullet$–ϕ relation.

For the selection of our galaxies, we turned to the Third Reference Catalog of Bright Galaxies (de Vaucouleurs et al. 1991, hereafter RC3). The full RC3 sample contains 23,022 galaxies (17,801 with morphological classifications) with apparent diameters larger than $1'$ ($D_{25} > 1'$), total B-band magnitudes brighter than about 15.5 mag (Vega $B_T \lesssim 15.5$ mag), and recessional velocities $cz < 15{,}000\ \mathrm{km\ s^{-1}}$ ($z < 0.050$; luminosity distances out to $d = 230$ Mpc). Using their online database,[18] we selected a sample consisting of all Sd galaxies (Hubble sequence morphological stage, $T = 7.0$)[19] with inclination angles (θ) such that $0° \leqslant \theta \leqslant 30°$, and in the northern celestial hemisphere (declinations greater than 0°). Here, the inclination angle is determined from the mean ratio ($\log R_{25}$) of the major isophotal diameter (D_{25}) to the minor isophotal diameter (d_{25}) measured at or reduced to the B-band surface brightness level $\mu_B = 25.0\ \mathrm{mag\ arcsec^{-2}}$. Therefore, we selected $0 \leqslant \log R_{25} \lesssim 0.06$ because $\theta \equiv \sec^{-1}(R_{25})$. These selection criteria yielded a sample of 85 spiral galaxies (see Table 1). We chose to restrict our sample to the northern sky so that our entire sample would be visible by extensive sky surveys such as Pan-STARRS1 and Sloan Digital Sky Survey (SDSS) to assure uniform access to high-quality imaging of volute structure to facilitate the measurement of spiral-arm pitch angles. Therefore, our sample should represent roughly 1/15 of the entire RC3 sample of Sd galaxies (i.e., half of the sky and $1 - \cos 30°$ of all random inclination orientations). In actuality, there are 787 Sd galaxies in the RC3; thus we have selected $85/787 \approx 11\%$, see $1/15 \approx 7\%$.

3. Predicting Black Hole Masses

3.1. The $M_\bullet$–ϕ Relation

The $M_\bullet$–ϕ relation is the most natural black hole mass scaling relation for application to spiral galaxies. Furthermore, it exhibits the lowest level of intrinsic scatter ($\epsilon = 0.33 \pm 0.08$ dex from Davis et al. 2017) of any single black hole mass scaling relation for spiral galaxies. The logarithmic spiral pitch angle is the most widely adopted metric for quantifying the geometric shape of spiral arms in disk galaxies. The $M_\bullet$–ϕ relation (Seigar et al. 2008; Berrier et al. 2013; Davis et al. 2017) can be used to predict central black holes in spiral galaxies and identify IMBH candidates (Graham et al. 2019; Treuthardt et al. 2019). The $M_\bullet$–ϕ relation is uniquely applicable to only spiral galaxies. Whereas most other black hole mass scaling relations may be employed for any morphological type of galaxy, such scaling relations are usually less accurate when derived from spiral galaxies alone. For example, Sahu et al. (2019a) found that the $M_\bullet$–σ_0 relation for late-type galaxies exhibits an intrinsic scatter that is 0.25 dex higher than the $M_\bullet$–σ_0 relation for early-type galaxies.

We have measured ϕ for all 85 galaxies in our sample (column 6 of Table 1). We obtained g, r, i, z, and y images from Pan-STARRS1[20] (Chambers et al. 2016), and the image that best highlighted the spiral structure was adopted for the pitch angle measurement. SDSS or Galaxy Evolution Explorer

[13] Nevertheless, there is evidence that IMBHs may reside in irregular and/or Magellanic-type morphologies (e.g., NGC 5408; Strohmayer & Mushotzky 2009).

[14] Observed in the B band, the B/T for Sd galaxies has been reported as 0.029 (Simien & de Vaucouleurs 1986) and $0.027^{+0.066}_{-0.016}$ (Graham & Worley 2008).

[15] For Sd galaxies, which typically lack conventional bulges, the central component proxy is likely a weak pseudobulge or "barge" (portmanteau of "bar" and "bulge").

[16] See Bohn et al. (2020) for a thorough list and discussion of known bulgeless galaxies with black hole mass estimates.

[17] Moreover, the fact that black holes follow the (black hole mass)–(host stellar mass) relation even in bulgeless galaxies indicates that massive black holes may form before stellar bulges in galaxies (e.g., Chen et al. 2023c).

[18] https://heasarc.gsfc.nasa.gov/W3Browse/all/rc3.html

[19] In their study of spiral galaxies in the RC3, Ma et al. (1999) indeed found that Sd types had the highest observed pitch angles, $|\bar{\phi}| = 25\overset{\circ}{.}00$.

[20] https://ps1images.stsci.edu/cgi-bin/ps1cutouts

Table 1
Sample of 85 Sd Galaxies

Galaxy (1)	Distance (Mpc) (2)	BPT (3)	CXO Exp (ks) (4)	CXO # (5)	$\lvert\phi\rvert$ (°) (6)	σ_0 (km s^{-1}) (7)	$\mathcal{M}_{\mathrm{gal},\star}$ (dex) (8)	θ (°) (9)	$v_{\max}$ (km s^{-1}) (10)	$\mathcal{M}_\bullet$ (dex) (11)	$P(\mathcal{M}_\bullet \leqslant 5)$ (%) (12)	$n\sigma$ (σ) (13)
IC 951	62.2 ± 7.2	H II	⋯	⋯	9.5 ± 2.9	⋯	10.2 ± 0.2	17 ± 2 (0 ± 44)	199 ± 29	7.6 ± 0.4	≈0	−5.9
IC 1221	74.9 ± 6.5	H II	5	0	22.7 ± 3.9	⋯	10.3 ± 0.1	18 ± 1 (27 ± 13)	62 ± 5	4.2 ± 0.5	93.3	1.5
IC 1774	41.8 ± 5.8	⋯	⋯	⋯	22.8 ± 2.4	⋯	9.9 ± 0.2	43 ± 2 (21 ± 17)	99 ± 5	5.0 ± 0.4	49.7	0.0
IC 1776	40.9 ± 6.1	⋯	⋯	⋯	17.9 ± 3.0	⋯	9.9 ± 0.2	28 ± 1 (12 ± 30)	99 ± 3	5.6 ± 0.4	8.0	−1.4
NGC 2500	12.0 ± 9.4	⋯	3	7	16.3 ± 1.3	⋯	9.6 ± 0.7	28 ± 11 (24 ± 9)	90 ± 34	5.6 ± 0.7	20.7	−0.8
NGC 2657	59.0 ± 7.2	⋯	⋯	⋯	15.8 ± 4.5	⋯	10.4 ± 0.2	35 ± 3 (12 ± 36)	144 ± 11	6.4 ± 0.5	0.4	−2.6
NGC 3906	19.0 ± 8.0	⋯	5	0	16.6 ± 1.9	⋯	9.6 ± 0.4	42 ± 4 (27 ± 10)	134 ± 12	6.2 ± 0.3	≈0	−3.6
NGC 3913	17.9 ± 7.5	⋯	5	0	12.9 ± 0.2	⋯	9.5 ± 0.4	16 ± 4 (12 ± 18)	69 ± 19	5.5 ± 0.5	19.0	−0.9
NGC 4393	7.7 ± 1.3	⋯	5	0	9.7 ± 2.8	⋯	8.6 ± 0.2	56 ± 1 (21 ± 17)	62 ± 2	5.6 ± 0.4	4.7	−1.7
NGC 5148	87.8 ± 6.5	⋯	7	0	18.7 ± 3.7	⋯	10.1 ± 0.1	25 ± 1 (21 ± 17)	84 ± 6	5.2 ± 0.5	34.5	−0.4
NGC 5668	21.3 ± 6.2	⋯	⋯	⋯	27.3 ± 2.8	53 ± 8	10.1 ± 0.3	15 ± 1 (24 ± 9)	152 ± 8	5.2 ± 0.4	31.2	−0.5
NGC 6617	91.4 ± 6.5	⋯	⋯	⋯	13.3 ± 1.9	⋯	10.2 ± 0.2	52 ± 1 (29 ± 12)	144 ± 6	6.7 ± 0.3	≈0	−5.5
NGC 6687	44.3 ± 6.0	⋯	⋯	⋯	24.5 ± 0.6	⋯	9.8 ± 0.2	42 ± 0 (12 ± 36)	101 ± 19[a]	4.9 ± 0.4	63.5	0.3
NGC 7363	91.2 ± 6.6	⋯	⋯	⋯	10.8 ± 1.3	⋯	10.5 ± 0.2	29 ± 1 (21 ± 21)	242 ± 14	7.8 ± 0.3	≈0	−10.1
NGC 7437	26.0 ± 5.9	⋯	⋯	⋯	18.0 ± 1.4	⋯	9.6 ± 0.3	17 ± 1 (0 ± 44)	153 ± 13	6.3 ± 0.3	≈0	−4.2
NGC 7535	58.5 ± 6.2	⋯	⋯	⋯	13.2 ± 1.0	⋯	10.2 ± 0.1	28 ± 2 (0 ± 44)	109 ± 8	6.2 ± 0.3	≈0	−4.3
UGC 42	69.2 ± 6.3	⋯	⋯	⋯	16.0 ± 4.8	⋯	9.5 ± 0.2	41 ± 6 (0 ± 104)	95 ± 11	5.7 ± 0.6	12.2	−1.2
UGC 283	47.5 ± 6.0	⋯	⋯	⋯	12.6 ± 4.0	⋯	9.8 ± 0.2	45 ± 1 (21 ± 17)	144 ± 9	6.8 ± 0.5	≈0	−3.6
UGC 336	75.0 ± 6.5	⋯	⋯	⋯	16.4 ± 3.1	⋯	10.0 ± 0.2	35 ± 3 (21 ± 14)	180 ± 16	6.7 ± 0.4	≈0	−4.1
UGC 384	50.9 ± 5.3	⋯	⋯	⋯	19.0 ± 0.7	⋯	10.0 ± 0.2	45 ± 2 (21 ± 17)	121 ± 6	5.8 ± 0.3	0.1	−3.1
UGC 1341	142.2 ± 6.6	⋯	⋯	⋯	10.8 ± 2.8	⋯	10.5 ± 0.1	32 ± 3 (21 ± 17)	208 ± 18	7.6 ± 0.4	≈0	−6.6
UGC 1544	55.5 ± 5.4	⋯	⋯	⋯	33.1 ± 1.9	⋯	9.6 ± 0.2	53 ± 7 (29 ± 14)	49 ± 5	2.5 ± 0.4	≈100	6.3
UGC 1606	104.9 ± 6.6	⋯	⋯	⋯	11.4 ± 4.7	⋯	9.9 ± 0.2	39 ± 2 (24 ± 18)	152 ± 8	7.0 ± 0.5	≈0	−3.7
UGC 1702	163.5 ± 6.6	⋯	⋯	⋯	5.8 ± 2.8	⋯	10.6 ± 0.2	20 ± 1 (21 ± 17)	284 ± 14	8.6 ± 0.4	≈0	−9.8
UGC 1795	187.1 ± 6.6	⋯	⋯	⋯	23.3 ± 3.6	⋯	10.6 ± 0.2	59 ± 3 (27 ± 31)	226 ± 18	6.4 ± 0.5	0.3	−2.8
UGC 1833	52.1 ± 5.4	⋯	⋯	⋯	25.3 ± 2.1	⋯	9.9 ± 0.2	61 ± 3 (21 ± 21)	82 ± 4	4.4 ± 0.3	95.6	1.7
UGC 1897	109.0 ± 6.6	⋯	⋯	⋯	28.7 ± 4.9	⋯	10.2 ± 0.2	40 ± 1 (27 ± 13)	132 ± 6	4.8 ± 0.7	61.8	0.3
UGC 2008	55.8 ± 5.5	⋯	⋯	⋯	21.6 ± 3.5	⋯	9.4 ± 0.2	28 ± 3(0 ± 61)	109 ± 12	5.3 ± 0.5	26.3	−0.6
UGC 2109	39.6 ± 5.3	⋯	⋯	⋯	17.0 ± 2.0	⋯	10.0 ± 0.2	39 ± 2 (29 ± 9)	112 ± 5	5.9 ± 0.3	0.4	−2.7
UGC 2437	40.1 ± 5.9	⋯	⋯	⋯	31.8 ± 3.7	⋯	10.0 ± 0.2	39 ± 2 (24 ± 18)	131 ± 9	4.4 ± 0.6	86.4	1.1
UGC 2458	218.3 ± 6.6	⋯	⋯	⋯	11.9 ± 0.9	⋯	10.7 ± 0.2	36 ± 2 (24 ± 35)	175 ± 34[a]	7.2 ± 0.4	≈0	−5.2
UGC 2623	47.5 ± 5.3	⋯	⋯	⋯	11.6 ± 3.0	⋯	9.8 ± 0.2	35 ± 4 (24 ± 18)	81 ± 9	5.9 ± 0.4	2.1	−2.0
UGC 2671	90.6 ± 6.4	⋯	⋯	⋯	16.6 ± 1.7	⋯	10.1 ± 0.2	25 ± 4 (0 ± 104)	187 ± 31	6.8 ± 0.4	≈0	−4.4
UGC 2935	56.8 ± 6.2	⋯	⋯	⋯	24.1 ± 2.9	⋯	9.1 ± 0.2	62 ± 2 (0 ± 53)	33 ± 2	3.0 ± 0.4	≈100	4.9
UGC 3074	52.7 ± 6.1	⋯	⋯	⋯	17.3 ± 0.2	⋯	9.9 ± 0.2	23 ± 4 (27 ± 13)	121 ± 22	6.0 ± 0.4	0.6	−2.5
UGC 3364	59.2 ± 6.4	⋯	⋯	⋯	16.5 ± 2.7	⋯	9.7 ± 0.2	33 ± 3(0 ± 61)	109 ± 9	5.9 ± 0.4	1.3	−2.2
UGC 3402	222.9 ± 6.6	⋯	⋯	⋯	16.6 ± 1.3	⋯	10.9 ± 0.2	34 ± 1 (27 ± 13)	202 ± 40[a]	6.9 ± 0.4	≈0	−4.5
UGC 3702	65.8 ± 6.6	⋯	⋯	⋯	16.2 ± 2.8	⋯	10.0 ± 0.2	27 ± 1 (0 ± 53)	65 ± 4	5.0 ± 0.4	48.0	0.0
UGC 3799	81.5 ± 6.7	⋯	⋯	⋯	10.7 ± 1.4	⋯	10.4 ± 0.2	39 ± 1 (27 ± 13)	144 ± 5	6.9 ± 0.3	≈0	−7.0
UGC 3826	26.8 ± 7.0	⋯	⋯	⋯	19.9 ± 4.8	⋯	9.3 ± 0.3	40 ± 4 (29 ± 9)	32 ± 3	3.4 ± 0.6	99.6	2.7
UGC 3875	71.6 ± 6.7	⋯	⋯	⋯	21.4 ± 3.8	⋯	10.0 ± 0.2	30 ± 5 (29 ± 12)	171 ± 25	6.1 ± 0.5	2.0	−2.0
UGC 3949	89.9 ± 6.9	⋯	⋯	⋯	25.8 ± 0.3	70 ± 5[b]	10.5 ± 0.1	32 ± 1 (0 ± 44)	101 ± 6	4.7 ± 0.3	87.6	1.2
UGC 4077	62.0 ± 6.7	H II	⋯	⋯	31.1 ± 1.1	⋯	10.0 ± 0.2	33 ± 3 (17 ± 25)	130 ± 12	4.5 ± 0.3	96.0	1.7
UGC 4363	48.8 ± 6.5	⋯	⋯	⋯	17.6 ± 4.2	⋯	9.8 ± 0.1	76 ± 1 (12 ± 24)	36 ± 2	3.8 ± 0.5	98.7	2.2
UGC 4622	175.9 ± 6.8	Comp	⋯	⋯	23.4 ± 2.3	107 ± 8[b]	10.9 ± 0.1	19 ± 1 (21 ± 21)	201 ± 34[a]	6.2 ± 0.5	0.6	−2.5
UGC 4831	62.2 ± 7.0	H II	⋯	⋯	15.9 ± 0.7	⋯	9.8 ± 0.2	26 ± 3 (0 ± 44)	127 ± 16	6.2 ± 0.3	≈0	−3.7
UGC 5142	98.1 ± 7.3	⋯	⋯	⋯	9.9 ± 2.6	⋯	10.1 ± 0.2	36 ± 3 (0 ± 104)	103 ± 8	6.5 ± 0.4	≈0	−3.9
UGC 5344	62.0 ± 7.5	H II	⋯	⋯	15.3 ± 3.4	⋯	9.9 ± 0.2	28 ± 5 (17 ± 21)	37 ± 6	4.1 ± 0.5	95.7	1.7
UGC 5460	20.6 ± 8.0	⋯	15	0	26.2 ± 2.2	⋯	9.4 ± 0.4	54 ± 4 (17 ± 21)	46 ± 3	3.3 ± 0.4	≈100	4.6
UGC 6505	97.6 ± 6.8	⋯	⋯	⋯	10.0 ± 4.7	⋯	10.0 ± 0.2	19 ± 4 (17 ± 21)	115 ± 21[a]	6.6 ± 0.6	0.4	−2.7
UGC 6616	20.5 ± 7.4	⋯	⋯	⋯	28.6 ± 4.3	⋯	9.3 ± 0.4	30 ± 1 (17 ± 25)	90 ± 5	4.2 ± 0.6	92.3	1.4
UGC 6893	84.2 ± 6.6	⋯	⋯	⋯	15.9 ± 4.1	⋯	10.0 ± 0.1	17 ± 8 (24 ± 15)	187 ± 80	6.9 ± 0.9	1.7	−2.1
UGC 7942	7.0 ± 3.7	⋯	⋯	⋯	16.5 ± 1.3	⋯	8.1 ± 0.5	31 ± 14 (21 ± 10)	35 ± 12[a]	3.9 ± 0.6	95.7	1.7

Table 1
(Continued)

Galaxy (1)	Distance (Mpc) (2)	BPT (3)	CXO Exp (ks) (4)	CXO # (5)	$\lvert\phi\rvert$ (°) (6)	σ_0 (km s^{-1}) (7)	$\mathcal{M}_{\mathrm{gal},\star}$ (dex) (8)	θ (°) (9)	$v_{\max}$ (km s^{-1}) (10)	$\mathcal{M}_\bullet$ (dex) (11)	$P(\mathcal{M}_\bullet \leqslant 5)$ (%) (12)	$n\sigma$ (σ) (13)
UGC 8153	42.5 ± 7.2	H II	⋯	⋯	31.6 ± 3.5	⋯	9.8 ± 0.2	29 ± 3 (27 ± 13)	111 ± 9	4.1 ± 0.5	94.4	1.6
UGC 8171	292.5 ± 7.4	AGN	⋯	⋯	22.7 ± 1.3	⋯	11.2 ± 0.1	25 ± 4 (21 ± 17)	240 ± 40[a]	6.6 ± 0.4	≈0	−3.9
UGC 8436	45.8 ± 7.3	H II	⋯	⋯	10.6 ± 1.2	⋯	9.6 ± 0.2	24 ± 8 (21 ± 17)	71 ± 23	5.7 ± 0.6	11.4	−1.2
UGC 8611	41.4 ± 7.1	H II	⋯	⋯	16.4 ± 4.3	⋯	9.5 ± 0.2	43 ± 4 (27 ± 13)	102 ± 8	5.8 ± 0.5	7.6	−1.4
UGC 8637	90.7 ± 6.4	H II	⋯	⋯	24.7 ± 3.3	⋯	10.3 ± 0.1	29 ± 7 (21 ± 17)	97 ± 21	4.8 ± 0.6	65.5	0.4
UGC 8670	271.0 ± 7.4	⋯	⋯	⋯	20.9 ± 3.3	⋯	11.1 ± 0.1	18 ± 7(0 ± 44)	238 ± 39[a]	6.7 ± 0.5	≈0	−3.4
UGC 9008	72.3 ± 6.6	H II	⋯	⋯	11.5 ± 4.6	⋯	9.9 ± 0.1	25 ± 5 (0 ± 53)	130 ± 27	6.7 ± 0.6	0.4	−2.7
UGC 9010	98.6 ± 6.5	⋯	⋯	⋯	13.6 ± 0.9	⋯	10.1 ± 0.2	46 ± 3 (29 ± 12)	150 ± 11	6.7 ± 0.3	≈0	−6.3
UGC 9042	108.9 ± 6.5	H II	⋯	⋯	11.0 ± 2.8	⋯	9.9 ± 0.2	20 ± 5 (29 ± 12)	229 ± 59	7.7 ± 0.6	≈0	−4.7
UGC 9052	99.2 ± 6.8	⋯	⋯	⋯	15.4 ± 1.5	⋯	10.3 ± 0.2	28 ± 6 (0 ± 44)	137 ± 24[a]	6.4 ± 0.4	≈0	−3.4
UGC 9340	62.9 ± 6.6	H II	5	0	19.6 ± 2.4	⋯	9.9 ± 0.2	23 ± 10 (21 ± 14)	144 ± 60	6.0 ± 0.8	9.9	−1.3
UGC 9722	95.9 ± 8.0	⋯	⋯	⋯	6.2 ± 4.7	⋯	10.0 ± 0.2	29 ± 6 (29 ± 12)	113 ± 23[a]	7.0 ± 0.6	0.1	−3.2
UGC 10020	31.5 ± 6.8	H II	⋯	⋯	19.2 ± 3.5	⋯	9.6 ± 0.2	19 ± 2 (12 ± 30)	77 ± 7	5.0 ± 0.5	52.0	0.1
UGC 10146	102.2 ± 6.1	⋯	⋯	⋯	14.2 ± 2.9	⋯	10.1 ± 0.2	34 ± 1 (12 ± 36)	127 ± 6	6.4 ± 0.4	≈0	−3.6
UGC 10440	59.1 ± 6.4	⋯	⋯	⋯	34.5 ± 1.9	⋯	9.6 ± 0.2	33 ± 7 (17 ± 21)	68 ± 13	2.9 ± 0.5	≈100	4.4
UGC 10831	102.2 ± 6.1	⋯	⋯	⋯	20.2 ± 3.5	⋯	10.4 ± 0.1	41 ± 2 (27 ± 13)	275 ± 18	7.1 ± 0.5	≈0	−4.5
UGC 10922	113.5 ± 6.4	⋯	⋯	⋯	16.8 ± 0.8	⋯	10.2 ± 0.2	25 ± 5 (21 ± 21)	118 ± 24	6.0 ± 0.4	1.1	−2.3
UGC 11029	40.4 ± 5.9	⋯	⋯	⋯	24.3 ± 4.8	⋯	9.6 ± 0.2	39 ± 0 (0 ± 44)	123 ± 5	5.2 ± 0.6	36.5	−0.3
UGC 11113	31.4 ± 6.1	⋯	⋯	⋯	19.7 ± 2.9	⋯	9.4 ± 0.2	54 ± 1 (24 ± 15)	55 ± 2	4.3 ± 0.4	95.2	1.7
UGC 11386	102.5 ± 6.6	⋯	⋯	⋯	12.6 ± 2.7	⋯	10.2 ± 0.2	27 ± 3 (21 ± 41)	146 ± 19	6.8 ± 0.4	≈0	−4.2
UGC 11515	41.3 ± 30.6	⋯	⋯	⋯	25.4 ± 0.3	⋯	10.1 ± 0.7	36 ± 2 (0 ± 44)	109 ± 11	4.9 ± 0.3	66.6	0.4
UGC 11556	71.0 ± 6.2	⋯	⋯	⋯	23.8 ± 3.4	⋯	9.5 ± 0.2	30 ± 0 (0 ± 53)	79 ± 3	4.5 ± 0.5	85.3	1.1
UGC 11653	47.6 ± 5.5	⋯	⋯	⋯	17.5 ± 2.2	⋯	9.7 ± 0.2	42 ± 3 (17 ± 25)	108 ± 7	5.8 ± 0.3	1.5	−2.2
UGC 11699	58.4 ± 5.9	⋯	⋯	⋯	15.0 ± 1.3	⋯	9.8 ± 0.2	27 ± 4 (21 ± 17)	103 ± 14	5.9 ± 0.4	0.4	−2.7
UGC 11728	108.1 ± 6.6	⋯	⋯	⋯	11.5 ± 3.2	⋯	10.4 ± 0.1	22 ± 3 (29 ± 14)	103 ± 13	6.3 ± 0.5	0.2	−2.9
UGC 11992	42.8 ± 5.8	⋯	⋯	⋯	9.2 ± 2.0	⋯	9.4 ± 0.2	44 ± 2 (24 ± 35)	103 ± 4	6.5 ± 0.3	≈0	−5.0
UGC 12008	104.9 ± 6.8	⋯	⋯	⋯	11.0 ± 4.5	⋯	10.3 ± 0.2	31 ± 3 (27 ± 16)	210 ± 22	7.6 ± 0.5	≈0	−4.8
UGC 12015	100.4 ± 6.4	⋯	⋯	⋯	12.0 ± 4.7	⋯	10.2 ± 0.2	40 ± 2 (27 ± 10)	99 ± 7	6.2 ± 0.5	1.4	−2.2
UGC 12176	123.5 ± 6.7	⋯	⋯	⋯	15.4 ± 3.9	⋯	10.5 ± 0.2	41 ± 3 (27 ± 13)	52 ± 3	4.7 ± 0.5	73.1	0.6
UGC 12184	51.2 ± 6.1	⋯	⋯	⋯	16.0 ± 3.0	⋯	9.5 ± 0.1	19 ± 3 (0 ± 44)	90 ± 14	5.6 ± 0.5	9.8	−1.3
UGC 12289	135.8 ± 6.6	⋯	⋯	⋯	24.0 ± 4.7	⋯	10.5 ± 0.2	29 ± 8 (17 ± 21)	201 ± 53	6.1 ± 0.7	7.1	−1.5
UGC 12685	67.0 ± 6.3	⋯	⋯	⋯	20.7 ± 3.9	⋯	9.8 ± 0.2	28 ± 10 (17 ± 17)	125 ± 40	5.7 ± 0.7	19.0	−0.9

Notes. Column (1): galaxy name. Column (2): luminosity distance (in Mpc) from HyperLeda (Makarov et al. 2014). Galaxies closer than 200 Mpc have been adjusted according to the Cosmicflows-3 Distance–Velocity Calculator (Kourkchi et al. 2020), available at the Extragalactic Distance Database. Distances (d) less than 38 Mpc have computed expectation distances based on the smoothed velocity field from the numerical action methods model of Shaya et al. (2017). 38 Mpc $< d <$ 200 Mpc have computed expectation distances based on the smoothed velocity field from the linear density field model of Graziani et al. (2019). Redshift-dependent distances are calibrated to the Planck Collaboration et al. (2020) cosmographic parameters. Column (3): the BPT [N II]/Hα vs. [O III]/Hβ standard optical diagnostic classification. "H II" = H II-region-like galaxy, "AGN" = active galactic nucleus, and "Comp" = composite galaxy (likely to contain a metal-rich stellar population plus an AGN). Column (4): Chandra X-ray Observatory (CXO) exposure time (in kiloseconds) of the galaxy's nucleus. Column (5): the number of X-ray photons emanating from the galaxy's nucleus detected by the CXO. Column (6): absolute value of the face-on spiral-arm pitch angle (in degrees), derived primarily from Pan-STARRS1 imaging, and measured by the `2DFFT`, `SpArcFiRe`, and/or `Spirality` software packages. Column (7): central stellar velocity dispersion (in km s^{-1}) from HyperLeda. Column (8): logarithm of the total galaxy stellar mass (in $M_\odot$); we use the intrinsic K-band apparent magnitudes from HyperLeda, then converted to luminosity by correcting for distance, compensating for surface brightness dimming (Tolman 1930, 1934), and applying the solar absolute magnitude from Willmer (2018), which is multiplied by the mass-to-light ratio from (Davis et al. 2019b, Equation (8)). Column (9): inclination angle (in degrees), determined via our pitch angle measurement process. The parenthetical values are the inclination angles according to the RC3. Column (10): physical maximum velocity rotation (in kilometers per second) corrected for inclination, from HyperLeda. Column (11): predicted black hole mass from the planar relation of Davis & Jin (2023), i.e., combining columns (6) and (10). Column (12): probability that the central black hole is an IMBH, i.e., $P(\mathcal{M}_\bullet \leqslant 5)$ (via Davis & Graham 2021, Equation (7)). Column (13): number of standard deviations below $10^5\ M_\odot$, i.e., $n\,\sigma = (5 - \mathcal{M}_\bullet)/\delta\mathcal{M}_\bullet$.

[a] Velocity is estimated via the Tully–Fisher relation (Tully & Fisher 1977) as refined by Tiley et al. (2019).

[b] Obtained from the NASA-Sloan Atlas (http://www.nsatlas.org). The 1.″5-radius-aperture-based σ_0 value was normalized to match the 0.595 h^{-1} kpc-radius-aperture used in the homogenized system of HyperLeda via the prescriptions of Jorgensen et al. (1995).

(GALEX) images were also consulted and used if the volute structure was better resolved. Galaxy images were first deprojected to a face-on orientation and then analyzed by a combination of software packages, including 2DFFT (Davis et al. 2012, 2016), SpArcFiRe (Davis & Hayes 2014), and/or Spirality (Shields et al. 2015, 2022).[21]

3.2. The $M_\bullet$–v_{max} Relation

For most of our sample ($75/85 \approx 88\%$), we were able to obtain the physical maximum rotational velocities ($v_{\rm max} \equiv v_{\rm rot}$) from HyperLeda (Makarov et al. 2014). As for the remaining ten galaxies (see Table 1, column 10), we estimated the rotational velocities by applying the Tully–Fisher relation (Tully & Fisher 1977) as refined by Tiley et al. (2019). The HyperLeda velocities are derived from the original line-of-sight velocities (from both 21 cm line widths and/or rotation curves) and subsequently corrected for inclination. Although we used the inclination angles derived from the axial ratios taken from the RC3[22] to initially restrict our sample to only face-on ($\theta \leqslant 30°$) galaxies, we elected to use our inclination angles that were derived during the process of measuring pitch angle. We find an unremarkable agreement between our measured inclinations and those from the RC3, primarily because the latter have high levels of uncertainty. Indeed, the median agreement between the inclinations is 0.53σ (i.e., the joint number of standard deviations at which both observations agree) or approximately 60% mutual agreement. Most importantly, it was necessary to use our measured inclinations because 19 of the galaxies from the RC3 were listed as essentially face-on (i.e., $\theta \approx 0$), thus making rotational velocity measurements impossible, which is incompatible with the existence of line-of-sight velocities from HyperLeda.

In fact, the act of determining inclination angles via the process of measuring pitch angles and/or Fourier transforms of spiral galaxy images has shown itself to be an accurate alternative to traditional inclination measurements via isophotal analysis (e.g., Grosbøl 1985; Ma 2001; García-Gómez et al. 2004; Poltorak & Fridman 2007; Fridman & Poltorak 2010). Thus, reversing the procedure can yield precise inclination angle measurements of galaxies by adjusting the inclination angle until a galaxy's spiral structure is closely described by pure logarithmic spirals with ϕ such that the growth of the spirals (radius R as a function of galactocentric azimuth φ) is monotonic ($dR/d\varphi > 0$). As a result, some of our galaxies were subsequently found to have inclinations $>30°$, but we obtained more precise measurements. Our inclination estimates have a mean uncertainty of $3\overset{\circ}{.}4$, compared to $18\overset{\circ}{.}5$ for those from the RC3 axial ratios.[23] The $M_\bullet$–$v_{\rm max}$ and $M_\bullet$–ϕ relations are both determined from the same sample of galaxies, but the former exhibits a higher level of intrinsic scatter, $\epsilon = 0.45$ dex (Equation (10) from Davis et al. 2019c).

3.3. A Planar Black Hole Mass Scaling Relation

Ever since the Magorrian et al. (1998) relation came onto the scene, the breadth and variety of black hole mass scaling relations have grown at a seemingly exponential rate in astrophysical literature. In our larger study (Jin & Davis 2023), we endeavored to use modern machine learning methods to try and identify higher-dimensional black hole mass scaling relations (i.e., $M_\bullet$ plus two or more galactic parameters) that are more accurate predictors of black hole mass than some of the aforementioned two-dimensional black hole mass scaling relations (i.e., $M_\bullet$ plus *one* galactic parameter).[24] For that study, we utilized a subfield of machine learning called symbolic regression (Cranmer 2023) to search for the best mathematical expressions to fit our data set of dynamically measured SMBH masses and their host galaxy parameters. The parent study of Jin & Davis (2023) identified an ideal (i.e., optimally precise and simple) relationship among $M_\bullet$, ϕ, and $v_{\rm max}$ for spiral galaxies. We presented the details behind this trivariate relationship separately in Davis & Jin (2023).

From Davis & Jin (2023), the equation for the $M_\bullet$–ϕ–$v_{\rm max}$ relationship is

$$\mathcal{M}_\bullet = \alpha(\tan|\phi| - 0.24) + \beta \log\left(\frac{v_{\rm max}}{211\ {\rm km\ s^{-1}}}\right) + \gamma, \quad (1)$$

with $\alpha = -5.58 \pm 0.06$, $\beta = 3.96 \pm 0.06$, $\gamma = 7.33 \pm 0.05$, and $\epsilon = 0.22 \pm 0.06$ dex in the $\mathcal{M}_\bullet$-direction, with parameters identified by PySR[25] (Cranmer 2023) and refined via HyperFit[26] (Robotham & Obreschkow 2015, 2016). We utilize Equation (1) to predict black hole masses for all of our sample (see column 11 of Table 1). Figure 1 illustrates a plot of the plane and the location of all 85 of our galaxies on the plane. The orientation of the plane conforms with expectations—big black holes reside in galaxies that have *tightly* wound spiral arms and whose disks are *rapidly* rotating, and small black holes are found in galaxies that have *loosely* wound spiral arms and whose disks are *slowly* rotating. The combination of these parameters is not surprising; Sarkar et al. (2023) find that later morphological types are primarily flocculent (in contrast to grand-design) galaxies with more loosely wound spiral arms and slower rotational velocities. For additional analyses and discussions, see Davis & Jin (2023) regarding the planar relation for spiral galaxies and Jin & Davis (2023) for higher-dimensional relations featuring all galaxy types.

We present a couple of metrics to help codify our best IMBH-candidate host galaxies in the rightmost columns of Table 1. Column 12 displays the probability that the central black hole is an IMBH, i.e., $P(\mathcal{M}_\bullet \leqslant 5)$ (via Davis & Graham 2021, Equation (7)). Column 13 provides the number of standard deviations below $10^5\,M_\odot$, i.e., $n\,\sigma = (5 - \mathcal{M}_\bullet)/\delta\mathcal{M}_\bullet$, which is equivalent to $P(\mathcal{M}_\bullet \leqslant 5)$, e.g., $P(\mathcal{M}_\bullet \leqslant 5) = 68.3\% \equiv 1.0\sigma$. Together, we utilize these quality/qualifying parameters to categorize our sample into striations of likeliness that a galaxy is expected to harbor an IMBH.

Inclination (θ) is the one common source of error that affects both independent variables ($|\phi|$ and $v_{\rm max}$) in our study. Because of the small-angle approximation, an error in the inclination of a galaxy when it is close to face-on (0°) is more significant than an equally sized error when a galaxy is close to edge-on (90°). This heteroscedasticity negatively affects the calculation of the

[21] See Section 2.1 from Davis & Graham (2021) for further details.

[22] See the parenthetical values from column 9 of Table 1.

[23] From their study of uncertainties in the projection parameters of spiral galaxies, Barnes & Sellwood (2003) found systematic uncertainties of ≈4° in inclination derived from photometry due to the presence of nonaxisymmetric structures.

[24] Machine learning has also been utilized recently to try and identify globular clusters that host IMBHs (Pasquato et al. 2024).

[25] https://github.com/MilesCranmer/PySR/tree/v0.12.3

[26] https://github.com/CullanHowlett/HyperFit

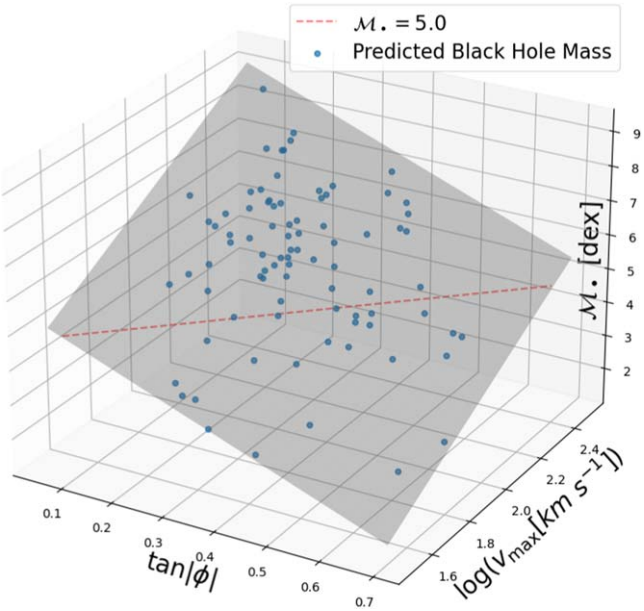

Figure 1. Here, we have reproduced the three-dimensional plot of the planar $M_\bullet$–ϕ–$v_{\max}$ relationship (Equation (1)) from Davis & Jin (2023), their Figure 2. Onto the surface (gray plane), we show the locations of all 85 of our galaxies from this work (blue dots) and a demarcation line (red dashed line) showing the boundary between SMBHs and IMBHs at $\mathcal{M}_\bullet = 5.0$. As shown, 23/85 (27%) of the galaxies lie below the dividing line, representing our IMBH-candidate host galaxies. Moreover, this enhanced plot illustrates that our galaxies are dispersed over the area of the plane, demonstrating a lack of degeneracy between the parameters by the apparent embedding of the two-dimensional manifold (i.e., surface) in three-dimensional space. The animation shows one rotation about the $M_\bullet$ axis. The real-time duration of the animation is 12 s. A copy is also available at the following link: http://surl.li/kkjeu.

(An animation of this figure is available in the online article.)

intrinsic $v_{\max}$ from the observed line-of-sight velocity ($v_{\rm LOS}$) because $v_{\max} \equiv v_{\rm LOS}\csc\theta$. However, the opposite effect is applicable to the measurement of pitch angles. Because a galaxy must be artificially projected into a face-on orientation to measure pitch angle, galaxies that are already close to a face-on orientation require minimal modification. Specifically, the minor-axis length (b) of a galaxy is stretched to equal its major-axis length (a), i.e., $a \equiv b \sec\theta$.

4. Results

4.1. Typical Central Black Hole in an Sd Galaxy

In Figure 2, we produce a histogram by summation of all 85 black hole mass estimates (determined by Equation (1)) for our sample of 85 Sd galaxies. We then fit a skew-kurtotic-normal distribution to the histogram. In doing so, we find that the typical Sd galaxy hosts a black hole with $\mathcal{M}_\bullet = 6.00 \pm 0.14$, with $P(\mathcal{M}_\bullet \leqslant 5) = 27.7\%$. Expressed differently, this most probable mass of $(1.01 \pm 0.33) \times 10^6\, M_\odot$ is $25.1\% \pm 8.2\%$ the mass of Sgr A* (Boehle et al. 2016).

Nominally, we expect that 1/3.61 of Sd galaxies harbor an IMBH. The RC3 provides morphological classifications for 17,801 diameter-selected galaxies, of which 787 are classified as Sd, i.e., 4.42%. A similar value of 3.9% (34/867) was found for Sd types by the diameter-selected sample of Lacerda et al. (2020). However, the magnitude-limited sample of the Carnegie-Irvine Galaxy Survey (Ho et al. 2011), plus the Milky Way, shows only 9/606 galaxies (1.49%) are type Sd, whereas its magnitude- and volume-limited subsample shows only 1/208 galaxies (0.5%) are type Sd (Davis et al. 2014; Mutlu-Pakdil et al. 2016). The EFIGI catalog (Baillard et al. 2011) is a subset of 4458 galaxies from the RC3, but no longer considers peculiar galaxies or galaxies with special features as belonging to separate stages. Thus, they classify an even higher fraction (285/4458 = 6.39%) of galaxies as $T = 7$ (de Lapparent et al. 2011).

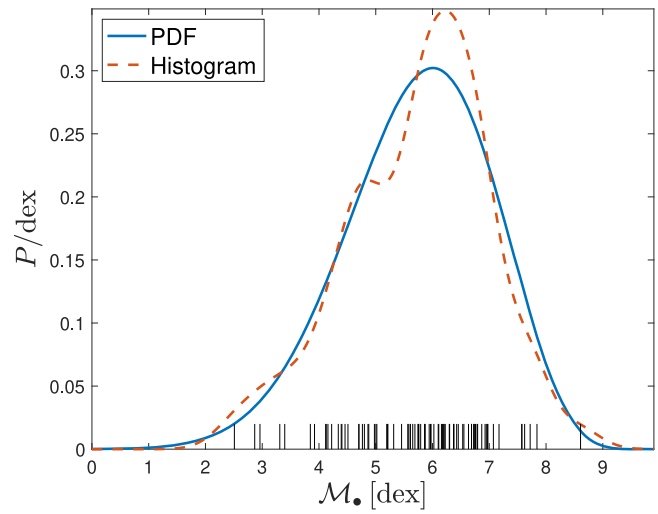

Figure 2. The distribution of black hole mass estimates ($\mathcal{M}_\bullet$) for the 85 Sd galaxies in our sample (from column 11 of Table 1) is shown as a rug plot along the bottom axis and has a mean $\mathcal{M}_\bullet$ of 5.73 ± 1.30 and a median of 5.89 ± 0.86. The smoothed histogram (red dashed line) is generated from the summation of all 85 mass estimates. The fitted PDF (blue solid line) for the Sd galaxies peaks at $\mathcal{M}_\bullet = 6.00 \pm 0.14$ (its mode). Here, the integrated probability under the PDF at $\mathcal{M}_\bullet \leqslant 5$ is 27.7%.

It is worth noting that there is an even higher fraction (i.e., almost double) of Sd galaxies in the nearby Universe; there are 39 Sd galaxies out of 454 galaxies with a classification (8.6%) in the RC3 at $cz < 1000\,\mathrm{km\,s^{-1}}$ ($z < 0.003$; luminosity distances out to $d = 15$ Mpc). This could indicate a real local excess or maybe a bias against Sd identification in more distant galaxies. Given this frequency of Sd galaxies in our local Universe, we would, therefore, expect that >218/17,801 or >1.22%[27] of bright galaxies ($B_{\rm T} \lesssim 15.5$ mag) in our local Universe host an IMBH, i.e., the "occupation fraction" (Zhang et al. 2009; Miller et al. 2015; Gallo & Sesana 2019). Using the cosmographic parameters from Planck Collaboration et al. (2020), this gives a comoving volume of 4.40×10^7 Mpc^3 for the RC3 extent and a lower limit for the number density of central IMBHs in our local Universe $>4.96 \times 10^{-6}$ Mpc^{-3}.

4.2. Additional Observational Data

4.2.1. BPT Classifications

We consulted the Reference Catalog of galaxy SEDs (RCSED; Chilingarian et al. 2017) to obtain nuclear classifications for each galaxy. The RCSED database utilizes the Baldwin–Phillips–Terlevich (hereafter BPT; Baldwin et al. 1981) [N II]/Hα versus [O III]/Hβ standard optical diagnostic classification, according to Kewley et al. (2006). As Birchall et al. (2022) point out, the BPT diagnostic clearly fails at reliably identifying sources as AGNs in lower-mass galaxies. Moreover, emission line flux ratio diagnostics can fail to

[27] This is a lower limit because we are not accounting here for other potential hosts of IMBHs (e.g., dwarf galaxies and globular clusters) or their frequency (albeit to a lesser extent) in earlier-type spirals.

identify entire subpopulations of AGNs when applied to single-fiber optical spectra (Comerford et al. 2022).

Other diagnostics, such as He II, may be better suited to detecting ionizing photons from AGNs in low-metallicity, star-forming dwarf galaxies (Umeda et al. 2022; Tozzi et al. 2023). As such, galaxies are classified as containing an H II-region, AGN, or both (composite). Only one galaxy (UGC 8171) is classified as having a bona fide AGN,[28] and another is a composite (UGC 4622). Coincidentally, both galaxies are predicted to have similar black hole masses (UGC 4622: $\mathcal{M}_\bullet = 6.16 \pm 0.46$, with $P(\mathcal{M}_\bullet \leqslant 5) = 0.6\%$; UGC 8171: $\mathcal{M}_\bullet = 6.55 \pm 0.39$, with $P(\mathcal{M}_\bullet \leqslant 5) \approx 0\%$), which are strongly inconsistent with being IMBHs.

Fourteen galaxies (including the composite UGC 4622) are classified as H II-region-like galaxies. The sample peak probability (mode) black hole mass for these 14 galaxies is $\mathcal{M}_\bullet = 5.14 \pm 0.35$, with $P(\mathcal{M}_\bullet \leqslant 5) = 36.0\%$. Therefore, these 14 H II-region-like galaxies have a slightly higher probability of containing central IMBHs than the typical Sd galaxy. A possible explanation for this difference could emerge from varied gas fractions between these subpopulations of galaxies (see a further discussion in Section 5.1).

4.2.2. Nuclear X-Ray Point Sources

We checked the CXO archive for observations of all galaxies in our sample. The CXO has observed eight of our galaxies; however, only one galaxy (NGC 2500) exhibited a measurable flux of X-ray photons emanating from its nucleus. With $\mathcal{M}_\bullet = 5.57 \pm 0.70$ and $P(\mathcal{M}_\bullet \leqslant 5) = 20.7\%$, NGC 2500 has a slightly higher probability of possessing an IMBH than the typical Sd galaxy. The seven galaxies without a measurable nuclear X-ray source have a sample mode black hole mass of $\mathcal{M}_\bullet = 5.79 \pm 0.41$, but $P(\mathcal{M}_\bullet \leqslant 5) = 41.4\%$ with a strong tail toward lower masses. Thus, these seven galaxies, where X-ray sources have been sought (but not found), are exactly *twice* as likely to harbor an IMBH in their nuclei (as compared to NGC 2500). Put another way, finding a nuclear X-ray source implies that a galaxy is twice as likely to harbor an SMBH, instead of an IMBH. Therefore, nondetection of nuclear X-ray sources could be a useful diagnostic for identifying IMBH host galaxy candidates.

This paltry harvest of nuclear X-ray sources for our sample is not unexpected, because we deliberately selected intrinsically faint Sd galaxies. With average black hole masses less than a million solar masses and Eddington ratios of $\sim 10^{-6}$, the typical Sd galaxy will likely have weak or no AGN signature. Also, Sd galaxies presumably lack one alleged mechanism behind AGN triggering, i.e., major mergers. Models suggest that major mergers may establish galaxy-wide, gravitationally induced torques that drive gas toward a galactic center, which may set off an AGN (e.g., Hopkins et al. 2006). However, the jury is decidedly hung when it comes to reaching a verdict about whether major mergers are relevant (e.g., Koss et al. 2010; Ellison et al. 2011; Hong et al. 2015; Weston et al. 2017; Goulding et al. 2018; Gao et al. 2020; Hernández-Toledo et al. 2023; Breiding et al. 2024) or not (e.g., Cisternas et al. 2011; Kocevski et al. 2012; Karouzos et al. 2014; Villforth et al. 2019; Lambrides et al. 2021) in triggering AGN. Optical emission line diagnostic tests for activity and/or X-ray emission from nuclei are usually reliable indications for the presence of an accreting SMBH. However, the lack of such indicators from the centers of Sd galaxies is not evidence for the lack of black holes, which may be in a quiescent phase of their duty cycles.

The occupation fraction of galaxies with nuclear X-ray emission can be quite different, depending on the sample selection. For example, Williams et al. (2022) found X-ray emission coincident within 2″ of their optical galaxy centers in the majority of their galaxies ($150/213 = 70.4\%$). Whereas Nwaokoro et al. (2021) found X-ray sources within 3″ of their host galaxy centers in only a tiny fraction of their sample ($7/1200 = 5.83‰$). The difference being that the former sample (Williams et al. 2022) was constructed as a statistically complete sample of nearby galaxies, while the latter sample (Nwaokoro et al. 2021) is a sample of only low-mass galaxies ($M_{\rm gal,\star} \lesssim 10^{9.5}\,M_\odot$). Moreover, Ohlson et al. (2024) found that in low-mass late-type galaxies X-ray fractions are lower and positional offsets of X-ray detections from their galactic centers are higher (possibly due to increased astrometric uncertainty). Thus, in general, exceedingly few X-ray sources exist in low-mass galaxies.

Moreover, an IMBH might not reside at the center of a galaxy. Due to their diminished mass, IMBHs traverse a difficult path sinking to the centers of their host galaxies and are more prone to wander away (Binggeli et al. 2000; Barrows et al. 2019; Bellovary et al. 2019; Pfister et al. 2019; Mezcua & Domínguez Sánchez 2020; Reines et al. 2020; Bellovary et al. 2021; Ma et al. 2021; Ricarte et al. 2021; Di Cintio et al. 2023; Partmann et al. 2024) as compared to their more massive SMBH cousins. However, Chu et al. (2023) point out that off-center SMBHs are common in the brightest cluster galaxies due to numerous galaxy mergers in their history, in which an SMBH can become significantly kicked out of the galactic center via dynamical interactions. Such offset SMBHs could potentially be detected via distortions that they cause in gravitational lens galaxies (Piro et al. 2023; Giani et al. 2023). In particular, perhaps the best (and most studied) IMBH candidate is HLX-1, which is 3.7 kpc from the center of ESO 243-49 (Farrell et al. 2009; Soria et al. 2010; Webb et al. 2010; Soria et al. 2011; Farrell et al. 2012; Soria et al. 2012; Webb et al. 2012; Soria et al. 2013; Webb et al. 2017).[29] However, Weller et al. (2023b) and Di Matteo et al. (2023b) remark about the largely missing population of wandering IMBHs, which rarely reveal themselves in the hyperluminous X-ray sources (HLXs) regime. IMBHs may also be associated with the merging remnants of dwarf galaxies onto larger galaxies (Webb et al. 2010; Farrell et al. 2012; Mapelli et al. 2012; Soria et al. 2013; Kim et al. 2015; Mezcua et al. 2015; Kim et al. 2017; Webb et al. 2017; Kim et al. 2020; Graham et al. 2021a).

4.3. IMBH Targets of Interest

We define a target of interest to have $P(\mathcal{M}_\bullet \leqslant 5) > 50\%$ ($n\,\sigma > 0$). Looking down Table 1 column 12, we identify 23 targets of interest (points plotted below the dashed red line in Figure 1). Indeed, this fraction of galaxies ($23/85 = 27\%$) is comparable with the $P(\mathcal{M}_\bullet \leqslant 5) = 27.7\%$ value we obtain from Figure 2. For these 23 galaxies, we wager that the odds are favorable that they each host a central IMBH. Among these targets, UGC 1544 is our most likely example to host an IMBH,

[28] For comparison, Lacerda et al. (2020) found that none of the 34 Sd galaxies in their sample possess an AGN.

[29] See also the much-discussed ultraluminous source from a possible IMBH in Messier 82 (Patruno et al. 2006; Muxlow et al. 2010; Joseph et al. 2011; Pasham et al. 2014).

with an expected mass of only $323 \pm 293\,M_\odot$. Moreover, these 23 galaxies are all relatively nearby; their mean distance is 55.2 ± 29.2 Mpc, median distance is 52.5 ± 18.2 Mpc, and a peak probability distance of 42.1 ± 6.1 Mpc. Thus, these galaxies pose appealing targets for further study.

4.3.1. Targets with Additional Consideration

UGC 3826 possesses an NSC with a mass of $\mathcal{M}_{\rm NSC,\star} = 6.04 \pm 0.09$ (Georgiev & Böker 2014; Georgiev et al. 2016). By application of the $M_\bullet$–$M_{\rm NSC,\star}$ relation (Graham 2020, Equation (9)), we predict a black hole mass of $\mathcal{M}_\bullet = 3.53 \pm 0.95$, with $P(\mathcal{M}_\bullet \leqslant 5) = 93.9\%$. Additionally, we can apply the $M_\bullet$–$M_{\rm gal,\star}$ relation (Davis et al. 2018, Equation (3)) to the total stellar mass of UGC 3826 ($\mathcal{M}_{\rm gal,\star} = 9.33 \pm 0.27$) and predict a black hole mass of $\mathcal{M}_\bullet = 2.75 \pm 1.30$, with $P(\mathcal{M}_\bullet \leqslant 5) = 95.8\%$. As we can see, these black hole mass predictions are very consistent with the value we found from Equation (1), $\mathcal{M}_\bullet = 3.40 \pm 0.60$, with $P(\mathcal{M}_\bullet \leqslant 5) = 99.6\%$.

Additionally, we identified two of our galaxies with black hole mass estimates via the $M_\bullet$–$\mathcal{C}_{\rm FUV,tot}$ and $M_\bullet$–$\mathcal{C}_{\rm NUV,tot}$ relations of Dullo et al. (2020), where $\mathcal{C}_{\rm FUV,tot}$ is the total FUV $-[3.6\,\mu{\rm m}]$ color and $\mathcal{C}_{\rm NUV,tot}$ is the total NUV $-[3.6\,\mu{\rm m}]$ color. As expected, the $M_\bullet$–$\mathcal{C}$ relations from Dullo et al. (2020) are such that more massive black holes reside in redder galaxies. Similarly, Baker et al. (2023) found that more massive black holes reside in galaxies with higher metallicities. For UGC 8153, Dullo et al. (2020) predict $\mathcal{M}_\bullet = 4.89 \pm 0.85$, with $P(\mathcal{M}_\bullet \leqslant 5) = 55.1\%$, and $\mathcal{M}_\bullet = 4.59 \pm 0.85$, with $P(\mathcal{M}_\bullet \leqslant 5) = 68.5\%$, from $\mathcal{C}_{\rm FUV,tot}$ and $\mathcal{C}_{\rm NUV,tot}$, respectively. Also, we can apply the $M_\bullet$–$M_{\rm gal,\star}$ relation (Davis et al. 2018, Equation (3)) to the total stellar mass of UGC 8153 ($\mathcal{M}_{\rm gal,\star} = 9.78 \pm 0.16$)and predict a black hole mass of $\mathcal{M}_\bullet = 4.11 \pm 0.98$, with $P(\mathcal{M}_\bullet \leqslant 5) = 81.7\%$. Thus, the color-based and total stellar mass predictions for UGC 8153 are consistent with our predictions for an IMBH, $\mathcal{M}_\bullet = 4.13 \pm 0.55$, with $P(\mathcal{M}_\bullet \leqslant 5) = 94.4\%$.

Similarly, for UGC 10020, Dullo et al. (2020) predict $\mathcal{M}_\bullet = 5.46 \pm 0.85$, with $P(\mathcal{M}_\bullet \leqslant 5) = 29.4\%$, and $\mathcal{M}_\bullet = 5.35 \pm 0.85$, with $P(\mathcal{M}_\bullet \leqslant 5) = 34.1\%$, from $\mathcal{C}_{\rm FUV,tot}$ and $\mathcal{C}_{\rm NUV,tot}$, respectively. Furthermore, we can apply the $M_\bullet$–$M_{\rm gal,\star}$ relation (Davis et al. 2018, Equation (3)) to the total stellar mass of UGC 10020 ($\mathcal{M}_{\rm gal,\star} = 9.59 \pm 0.22$) and predict a black hole mass of $\mathcal{M}_\bullet = 3.55 \pm 1.14$, with $P(\mathcal{M}_\bullet \leqslant 5) = 89.9\%$. Likewise, the color-based total stellar mass predictions for UGC 10020 are consistent with our predictions for an IMBH, $\mathcal{M}_\bullet = 4.98 \pm 0.47$. However, the higher color-based mass predictions for UGC 10020 are in line with our weak $P(\mathcal{M}_\bullet \leqslant 5) = 52.0\%$, which is the least certain IMBH candidate among our sample of 23 targets of interest.

5. Discussion

5.1. The Prototypical Sd Galaxy

We observe that the geometry of the spiral-arm shape in Sd galaxies is not always as loosely wound as one might expect. Some of this is due to obvious misclassifications of the morphological type.[30] For example, our smallest pitch angle measurement is $|\phi| = 5\overset{\circ}{.}8 \pm 2\overset{\circ}{.}8$ for UGC 1702, which is an absurdly small pitch angle for a legitimate Sd galaxy. However, UGC 1702 is also one of the most massive galaxies in our sample, with $\mathcal{M}_{\rm gal,\star} = 10.55 \pm 0.21$. Moreover, it has a high rotational velocity with $v_{\rm max} = 283.8 \pm 14.3$ km s^{-1}. Indeed, our trivariate relation predicts the most massive SMBH in our sample, with a mass of $\mathcal{M}_\bullet = 8.61 \pm 0.37$, with $P(\mathcal{M}_\bullet \leqslant 5) \approx 0\%$.

In some instances, the geometry of arms may not agree well with a weak bulge. The RC3 bases their classifications on the de Vaucouleurs (1959) classification approach, which is based on the appearance of the spiral arms and the bulge of a galaxy. They consider both (*i*) the degree of openness (i.e., $|\phi|$) and (ii) the resolution of spiral arms into star clusters or very luminous stars (i.e., knotty versus smooth spiral structure), and additionally (iii) the relative prominence of the bulge or central concentration (i.e., B/T).[31] These criteria that govern the stage for spirals may be inconsistent in some cases or may be overruled by other factors that affect the morphological type (Sandage 1961; Sandage & Bedke 1994). Importantly for our study of Sd types, Mengistu & Masters (2023) find that bluer and lower-mass galaxies most closely follow the "expected" arm windiness correlation with bulge size, i.e., smaller bulges with loosely wound spiral arms.

The B/T ratio is directly related to the morphological type, in general, but with considerable scatter for a given type (Simien & de Vaucouleurs 1986; Laurikainen et al. 2007; Graham & Worley 2008; Willett et al. 2013). Specifically, Masters et al. (2019) find that galaxies with larger bulges favor tighter spiral arms, while those with smaller bulges have a wide range of arm winding; see Lingard et al. (2021), who find no correlation between bulge size and pitch angle. Very recently, Chugunov et al. (2024) affirmed "that the pitch angle of spiral arms decreases with increasing bulge or bar fraction." Smith et al. (2022) found that blue (i.e., star-forming) galaxies predominantly exhibit loosely wound spiral arms and red (i.e., quiescent) galaxies mainly display tightly wound spiral arms. Similarly, late-type spirals have been shown to have stronger arms (Yu & Ho 2020) and higher star formation rates (Yu et al. 2021).[32]

Importantly, multiple studies find a general trend (albeit with considerable scatter) between pitch angle and Hubble stage, i.e., $|\phi| \propto T$ (Kennicutt 1847; Seigar & James 1998; Ma et al. 1999; Baillard et al. 2011; Yu et al. 2018; Díaz-García et al. 2019; Yu & Ho 2019, 2020; Chugunov et al. 2024). More precisely, Treuthardt et al. (2012) show that the correlation between $|\phi|$ and T is tightest when selecting spiral galaxies with fast rotating bars, which is in close agreement with the theoretical relation (Roberts et al. 1975). Notably, Yu & Ho (2019) find that $|\phi|$ is most closely correlated with $M_{\rm gal,\star}$, especially so for low-mass galaxies. Overall, there are related connections between multiple parameters, with positive correlations between ϕ–T and ϕ–(absolute magnitude) relations

[30] Consulting the meta-analysis of morphological types for our galaxies from HyperLeda, we find that only two of our galaxies have morphological types that do not agree ($T \neq 7$) with our adopted classifications from the RC3: UGC 283, $T = 5.8 \pm 0.7$ and UGC 10146, $T = 8.7 \pm 1.4$.

[31] Willett et al. (2013) and Masters et al. (2019) argue that modern morphological classifications of spiral galaxies have devolved the traditional tenets of the classic Hubble-Jeans sequence (Jeans 1919; Lundmark 1925; Hubble 1926a, 1926b, 1927; Jeans 1928; Hubble 1936) that prioritized spiral-arm winding (van den Bergh 1998; de Lapparent et al. 2011), but still considered bulge size; contemporary morphological sequences are now predominantly ordered on central bulge size alone, with no reference to spiral arms (Graham & Worley 2008; Willett et al. 2013).

[32] Aktar et al. (2023) found no evidence of a trend between star formation rates and pitch angle, but postulated that the lack of a correlation "may be explained by different star formation efficiencies caused by the distinct galactic ambient conditions."

(Kennicutt 1847), and negative correlations between ϕ–$v_{\max}$ (Kennicutt 1847; Davis et al. 2019c) and $v_{\max}$–T (Roberts 1978) relations.

Frustratingly, if the image resolution is inadequate, the degree of openness cannot be accurately determined, and the default is toward a later type. Indeed, Peng et al. (2018) empirically found that $|\phi| \propto z$, that is, the average pitch angle observed in galaxies is perceived to increase (loosen) as a function of redshift. Although, this is not expected to be an intrinsic effect, but rather a systematic effect of "tightly wound arms becoming less visible as image quality degrades" (Peng et al. 2018). Hence, some of the more distant galaxies in our sample of "Sd" galaxies could have tightly wound spiral arms that were not accounted for in classification catalogs from decades ago due to lower image resolutions. From that point of view, the ($T = 7.0$) morphological types for our sample could be considered upper limits. Additionally, Graham & Worley (2008) point out that disk luminosities become progressively dimmer with increasing Hubble type, further exacerbating efforts to resolve the geometry of spiral arms. As telescope technology continues to advance, it becomes increasingly probable that new, high-resolution surveys may overturn old morphological classifications if they can better resolve the geometry of the spiral arms.

Even though the Sd sample tends toward large pitch angles, the broad range they cover could be a possible indication of their diverse origins or evolutionary histories. The gas content in Sd galaxies[33] could also have a role to play in diminished pitch angle values. Davis et al. (2015) observed that, due to spiral density wave theory (i.e., Lin & Shu 1966), the pitch angle of spiral galaxies is due to both the central mass of a galaxy (which includes the central black hole), as well as the density of the disk of a galaxy, such that the pitch angle is directly proportional to the disk density and inversely proportional to the central mass. Graham & Worley (2008) show that the surface density of disks decreases as the morphological type of a galaxy becomes increasingly later. This can become problematic in accounting for later morphological types in the galaxy stellar mass function; Kim et al. (2023) estimate that the majority of low-surface brightness galaxies are missed in redshift surveys at $z > 0.9$. Indeed, the problem is not easily ignorable since the number density of galaxies *increases* monotonically as their stellar masses decrease (Driver et al. 2022). Moreover, the prominence of low stellar-mass galaxies becomes increasingly relevant as the slope of the galaxy stellar mass function is shown to steepen with redshift (Navarro-Carrera et al. 2024).

Thus, if the ratio of disk density to central mass does not decrease at the same rate as the morphological type increases, the pitch angle of an Sd galaxy could very well be lower than expected if it were only due to the central mass. Another complication to consider is the environment of each galaxy. Cluster galaxies can experience ram pressure stripping, which "unwinds" their spiral arms (Bellhouse et al. 2021). Although, this effect would act to loosen (i.e., increase $|\phi|$) the pitch angle of a galaxy. Therefore, our pitch angle measurements could be skewing the predicted black hole masses toward higher masses in some cases, depending on the influence of unknown densities.

We are also cognizant of the possibility that other spiral genesis mechanisms besides the spiral density wave theory could be at play, e.g., swing amplification (see Dobbs & Baba 2014, for a review). Notably, Hart et al. (2018) found that ≈40% of the galaxies in their sample have spiral arms that can be modeled by swing amplification. Yu & Ho (2020) compared their work with the models of Hart et al. (2018) and speculated that later morphologically typed spiral galaxies in their sample could be similarly influenced by swing amplification. Following the work by Yu & Ho (2018) and Pringle & Dobbs (2019), Lingard et al. (2021) found that their sample of spiral galaxies could be explained by the evolution of transient/recurrent spirals via swing amplification that winds up over time, if pitch angles are sufficiently high. Similarly, Reshetnikov et al. (2022) showed from observations that $|\phi| \propto z$, i.e., pitch angles tend to decrease (wind-up) with time. Notably, Hart et al. (2017) found that central mass concentration alone does not govern pitch angle; they found that galaxies that are more disk dominated contain more spiral arms with tighter pitch angles.

It is also notable that there appears to be an empirical correlation between pitch angle and dark matter halos; Seigar et al. (2005, 2006, 2014) have demonstrated an anticorrelation between pitch angle and the central mass concentration of a spiral galaxy via measurement of the rate of shear of its rotation curve (Γ).[34] Additionally, simulations have shown that late-type bulgeless galaxies pose an enigma due to an apparent dichotomy between their observed and simulated angular momenta. D'Onghia & Burkert (2004) demonstrated that in the absence of major mergers, dark matter halos have too low an angular momentum to reconcile the observed disks of their embedded bulgeless late-type galaxies. Indeed, Rodriguez-Gomez et al. (2022) found that galaxies with higher specific angular momenta reside in faster-spinning halos and tend to host less massive black holes. Also, Rodriguez-Gomez et al. (2022) further described that halo spin is anti-correlated with black hole mass at a fixed galaxy or halo mass. Such complications could be an additional factor in discrepancies between our observed properties (i.e., pitch angle, stellar mass, and rotational velocities). This further warrants and justifies our adoption of a higher-dimensional black hole mass predictor, rather than relying on only one two-dimensional scaling relation parameterization (see also Williams et al. 2023b).

Overall, we find that the notion of a truly average Sd galaxy is false. Similarly, Daniels (1952) found, in his landmark anthropometric study, that the *average man* does not exist. From a study of 4063 men, Daniels (1952) found that no man was average (i.e., within ±0.3 standard deviations of the mean) across more than nine out of the 132 body measurements performed for the study. Using the same definition of average, we find that only three of our 85 galaxies are "average" across $|\phi|$, $v_{\max}$, and $\mathcal{M}_{\mathrm{gal},\star}$. These most average galaxies include UGC 384, 2109, and 3074. Together, these three prototypical Sd galaxies have a modal $\mathcal{M}_{\bullet} = 5.84 \pm 0.19$, with $P(\mathcal{M}_{\bullet} \leqslant 5) = 0.3\%$. Therefore, IMBHs are likely to be harbored only in Sd galaxies that are significantly looser wound, rotating slower, and/or less massive than the exemplar of Sd galaxies.

[33] Lacerda et al. (2020) found that Sd galaxies have some of the highest gas fractions (f_{gas}) of any morphological type (see their Figure 9 and Table 3), with an average $f_{\mathrm{gas}} = 6.76\%$, which is almost three times the average proportion they found in their Sa galaxies ($f_{\mathrm{gas}} = 2.40\%$).

[34] We note that subsequent studies (Kendall et al. 2015; Yu et al. 2018; Yu & Ho 2019) have also found an anticorrelation between $|\phi|$ and Γ, albeit a much weaker anticorrelation with higher scatter.

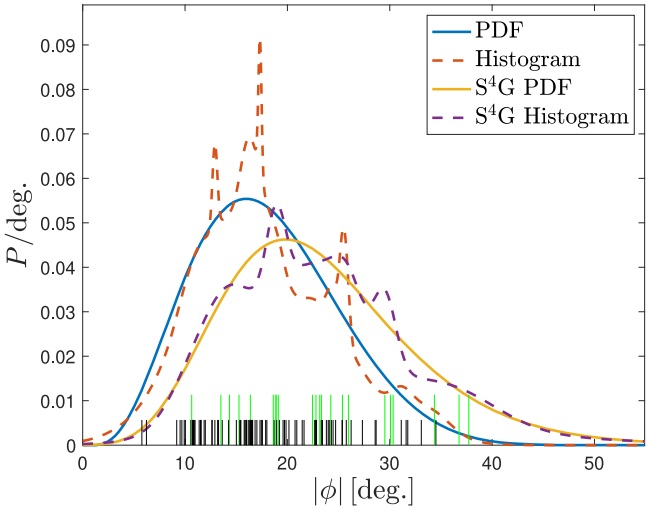

Figure 3. The distribution of the pitch angles ($|\phi|$) for our sample of 85 Sd galaxies (from Table 1, column 6) is shown as a rug plot with short black tassels with a mean $|\phi|$ of $18\overset{\circ}{.}0 \pm 7\overset{\circ}{.}0$ and a median of $17\overset{\circ}{.}1 \pm 4\overset{\circ}{.}7$. The smoothed histogram (red dashed line) is generated from the summation of all 85 pitch angle measurements. The fitted PDF (blue solid line) peaks at $|\phi| = 16\overset{\circ}{.}0 \pm 0\overset{\circ}{.}8$. We compare this with a sample of 22 Sd galaxies (purple dashed line) shown as a rug plot with long green tassels with pitch angle measurements from the S^{4}G sample (Herrera-Endoqui et al. 2015; Díaz-García et al. 2019), with a mean $|\phi|$ of $23\overset{\circ}{.}3 \pm 9\overset{\circ}{.}4$, a median of $22\overset{\circ}{.}2 \pm 6\overset{\circ}{.}0$, and a fitted PDF (yellow solid line) that peaks at $|\phi| = 19\overset{\circ}{.}8 \pm 2\overset{\circ}{.}0$.

5.2. Sd Classifications: Optical versus Mid-IR

In order to draw comparisons of our population of Sd galaxies, we sought an independent study that had a statistically viable number of Sd galaxies with quantitative pitch angle measurements. The largest such sample we could find comes from the Spitzer Survey of Stellar Structure in Galaxies (Sheth et al. 2010, hereafter S^{4}G). A series of papers (Buta et al. 2010, 2015) goes through careful morphological classifications of their 2352 galaxies. Using the classifications of Buta et al. (2015) and pitch angle measurements by Herrera-Endoqui et al. (2015), refined by Díaz-García et al. (2019), we identified 22 Sd galaxies with pitch angle measurements from the S^{4}G sample.

In Figure 3, we present a comparison of the pitch angle distribution of our 85 Sd galaxies alongside the 22 Sd galaxies from Díaz-García et al. (2019). Both distributions demonstrate a similar shape (i.e., positive skewness), but the peaks are notably different: $|\phi| = 16\overset{\circ}{.}0 \pm 0\overset{\circ}{.}8$ for our galaxies and $|\phi| = 19\overset{\circ}{.}8 \pm 2\overset{\circ}{.}0$ for the galaxies from the S^{4}G sample. We performed a Kolmogorov–Smirnov (K-S) test (Kolmogorov 1933; Smirnov 1948) to assess the likelihood that both samples come from the same distribution. In doing so, we found a p-value $= 0.0110$, thus rejecting the null hypothesis at the 1.10% level (i.e., a 98.9% probability that the samples come from different parent populations).

However, this is not the end of the story for this comparison. Crucially, our 85 galaxies were classified as Sd galaxies by the RC3 based on B-band images, whereas the 22 galaxies classified as Sd galaxies by Buta et al. (2015) were based on middle-infrared (mid-IR) images. Specifically, Buta et al. (2015) exclusively used 3.6 μm images, which highlight the photospheric light of old stars (Pahre et al. 2004). This can lead to a complicated difference in the perceived morphologies that were traditionally observed in B-band light that highlights young stellar populations and is strongly affected by extinction and reddening. Moreover, Buta et al. (2015) also used an updated morphological notation system (e.g., Buta 2014) that is similar to, but more extensive than, the notational system provided in the RC3.

Overall, it is generally accepted that the morphological classification schemes derived from B-band observations can be effectively applied to infrared images and reproduce the gamut of morphological diversity (Eskridge et al. 2002; Buta et al. 2010, 2015). However, it has been claimed that optical and near-IR morphologies of spiral galaxies are uncorrelated; Block & Puerari (1999) found that "the Hubble tuning fork does not constrain the morphology of the old stellar Population II disks." Moreover, Block & Puerari (1999) found extreme examples where galaxies on opposite ends of the Hubble tuning fork in optical light can have the same morphology when observed in the near-infrared (K' band). Nonetheless, Eskridge et al. (2002) found that B-band versus H-band morphologies are mostly similar. Separately, Buta et al. (2010, 2015) found that B-band versus 3.6 μm morphologies are also largely the same. Although, systematic differences are observed.[35]

Infrared morphologies exhibit a clear "earlier effect" (Eskridge et al. 2002; Buta et al. 2010, 2015), which is a result of the increased prominence of the bulge and the decreased prominence of star-forming regions in spiral arms when observed in the infrared. Indeed, the extensive multi-wavelength bulge-disk decomposition study of M81 (NGC 3031) by Gong et al. (2023) showed that the Sérsic index (Sersic 1968) and effective radius of its bulge are proportional to wavelength, so much so that M81 appears to have a prominently classical bulge in the infrared and bulgeless at ultraviolet wavelengths. However, Ito et al. (2024) saw a negative correlation between the observed wavelength and effective radius of $z \geqslant 3$ quiescent galaxies, and their resulting size–mass relation is lower than those observed at lower redshifts (e.g., van der Wel et al. 2014; Hon et al. 2022, 2023). See also Yao et al. (2023); Ono et al. (2024), and Ormerod et al. (2024), for their studies of the variation of morphological parameters with rest-frame wavelength. Thus, because the bulges visually stand out more in IR images, they tend to be classified as an earlier type than they were in B-band images. Specifically, Eskridge et al. (2002) find that their H-band classifications are systematically one stage earlier than the RC3 B-band classifications on average ($T_H = T_B - 1$). Therefore, it is plausible that the 22 S^{4}G $T_{3.6\,\mu\mathrm{m}} = 7$ galaxies that we compare with Figure 3 are actually $T_B = 8$ galaxies with expectedly higher pitch angles, whose classifications are being overruled by their more prominent bulges when observed in the mid-IR.

We identified one of the 22 S^{4}G galaxies (NGC 5668) also in our sample that was independently measured by Herrera-Endoqui et al. (2015) and Díaz-García et al. (2019). Díaz-García et al. (2019) measured a pitch angle of $|\phi| = 29\overset{\circ}{.}7 \pm 3\overset{\circ}{.}9$, which is slightly higher, but consistent with our determination of $|\phi| = 27\overset{\circ}{.}3 \pm 2\overset{\circ}{.}8$ for NGC 5668. However, Buta et al. (2015) classify NGC 5668 as SAB(rs)cd, or $T_{3.6\,\mu\mathrm{m}} = 6.5$. Thus, supporting our suspicions of the earlier effect in its classification, although

[35] For a further investigation of how pitch angle measurements vary as a function of the wavelength of light and comparisons with predictions of the density wave theory, see the following papers (e.g., Pour-Imani et al. 2016; Yu et al. 2018; Miller et al. 2019; Abdeen et al. 2020, 2022; Chen et al. 2023c; Martínez-García et al. 2023).

the pitch angles remain consistent in this case. When ultimately analyzed in aggregate, the statistically significant observation of larger absolute pitch angles for the 3.6 μm sample confirms the implications of our aforementioned K-S test that the samples are drawn from different populations. Verily, the comparison of morphologies across the electromagnetic spectrum is problematic and confounds demographical comparisons of galaxies.

5.3. Comparison with a General Spiral Population

In their study of a volume- and magnitude-limited sample of 140 spiral galaxies, Davis et al. (2014) found that the black hole mass function (BHMF) of spiral galaxies peaks at $1.17 \times 10^7\, M_\odot$. Their volume-limited sample of 140 spiral galaxies is comprised of Sa, Sab, Sb, Sbc, Sc, Scd, Sd, and Sm types. However, it included only one ($\approx$1%) Sd galaxy (ESO138-010); the most common morphological type was Sc (34%; see their Figure 3). For comparison with their BHMF, we have produced a distribution of our black hole mass estimates (Figure 2).[36] From Figure 2, we find that the black hole mass distribution of Sd galaxies is indeed significantly different from the general spiral galaxy sample of Davis et al. (2014, their Figure 7). Our fitted probability density function (PDF) peaks at $1.01 \times 10^6\, M_\odot$, or 8.6% of the most probable black hole mass found in an average (Sc) spiral galaxy.

We also present the distributions of pitch angles (Figure 3), rotational velocities (Figure 4), and total stellar masses (Figure 5). As we can see, these distributions exhibit slight telltale evidence of multiple populations also reflected by the multimodal distribution of predicted black hole masses in Figure 2. These similarities across independent measurements give credence to the seemingly disparate spiral geometries uncovered by our pitch angle measurements across this sample of only Sd galaxies.[37] We note that our general PDF fit to the pitch angle distribution in Figure 3 is not appreciably different from the general shape of the distribution for all spiral types (Davis et al. 2014, Figure 6).[38] However, the histogram for Sd galaxies in Figure 3 does exhibit a subpopulation of high pitch angle galaxies (e.g., the prominence at $|\phi| = 25\overset{\circ}{.}5$), and lacks the enhanced population of galaxies with $|\phi| \lesssim 10^\circ$ found in Davis et al. (2014). This result is also reflected in the recent work of Fusco et al. (2022), who conducted a follow-up study to Davis et al. (2014) by analyzing the complementary population of 74 low-mass galaxies (peak probability at $|\phi| = 17\overset{\circ}{.}5$) that was excluded from Davis et al. (2014). Fusco et al. (2022) report a similar enhancement to the BHMF from Scd-Sm galaxies, and show that galaxies in this morphological population are predominantly the hosts of "less-than-super-massive" black holes ($M_\bullet \lesssim 10^6\, M_\odot$).

5.4. Implications

5.4.1. Seeding Models

Astrophysicists are grappling with the puzzle of how SMBHs formed within the first billion years of the Universe's existence (at $z \gtrsim 6$). These SMBHs, some exceeding a billion solar masses (Tripodi et al. 2023), challenge conventional models of black hole formation from stellar collapse. Observations of quasars at high redshifts support the existence of these massive black holes (Fan et al. 2001, 2003; Willott et al. 2007; Mortlock et al. 2011; Morganson et al. 2012; Kashikawa et al. 2015; Wu et al. 2015; Bañados et al. 2016, 2018; Decarli et al. 2018; Matsuoka et al. 2019; Onoue et al. 2019; Endsley et al. 2023). The prevailing theory suggests that ancient "seeds" of black holes underwent rapid growth through accretion and mergers (e.g., Soltan 1982; Small & Blandford 1992; Kauffmann & Haehnelt 2000; Volonteri & Rees 2005; Shankar et al. 2009; Shen et al. 2020; Li et al. 2023; Lin et al. 2023). Theorists employ simulations and analytic

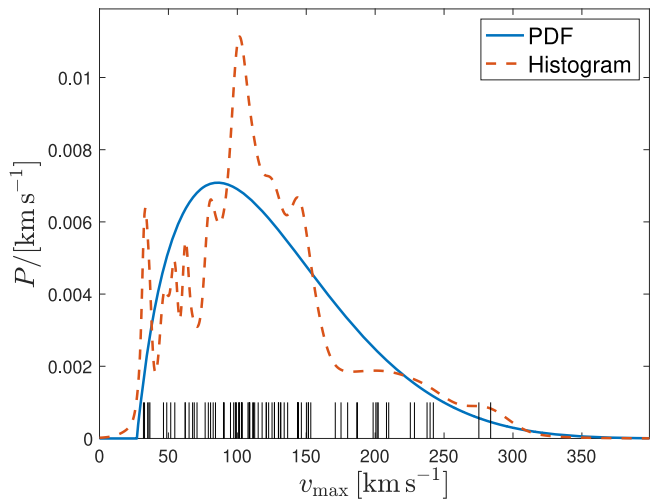

Figure 4. The distribution of the maximum rotational velocities ($v_{\max}$) for all 85 Sd galaxies (from Table 1, column 10) is shown as a rug plot along the bottom axis with a mean $v_{\max}$ of 124.9 ± 60.2 km s^{-1} and a median of 114.3 ± 32.9 km s^{-1}. The smoothed histogram (red dashed line) is generated from the summation of all 85 $v_{\max}$ measurements. The fitted PDF (blue solid line) peaks at $v_{\max} = 85.8 \pm 6.5$ km s^{-1}.

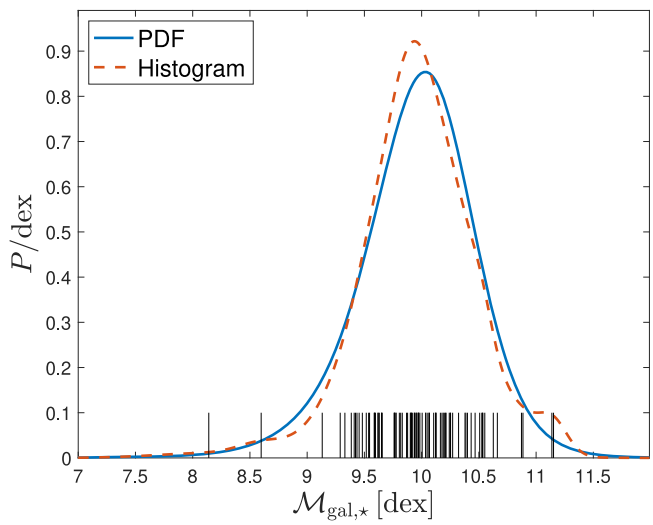

Figure 5. The distribution of the total galaxy stellar masses ($\mathcal{M}_{\mathrm{gal},\star}$) for all 85 Sd galaxies (from Table 1, column 8) is shown as a rug plot along the bottom axis with a mean $\mathcal{M}_{\mathrm{gal},\star}$ of 9.96 ± 0.53 and a median of 9.97 ± 0.30. For comparison, Lacerda et al. (2020) found an average value of $\mathcal{M}_{\mathrm{gal},\star} = 9.90 \pm 0.09$ from their sample of 34 Sd galaxies. The smoothed histogram (red dashed line) is generated from the summation of all 85 $\mathcal{M}_{\mathrm{gal},\star}$ measurements. The fitted PDF (blue solid line) peaks at $\mathcal{M}_{\mathrm{gal},\star} = 10.03 \pm 0.06$.

[36] In this and subsequent figures, the histograms are the summation of individual normal distributions that describe each measurement.

[37] See also the recent meta-analysis performed by Savchenko et al. (2020), whose Figure 13 demonstrates a significant variance of pitch angle across morphological types.

[38] The modes of both the distributions for Sd ($|\phi| = 16\overset{\circ}{.}0$) and all ($|\phi| = 18\overset{\circ}{.}5$) spiral galaxies are remarkably close to the pitch angle ($|\phi| \approx 17\overset{\circ}{.}0$) of the golden spiral (Davis et al. 2014, Appendix A).

studies to explore the possible scenarios leading to the formation of these massive black holes (Dijkstra et al. 2014; Visbal et al. 2014; Habouzit et al. 2016a, 2016b; Lupi et al. 2021; Bhowmick et al. 2022; Trinca et al. 2022; Jeon et al. 2023; Bennett et al. 2024), albeit with varying success rates (see Di Matteo et al. 2023a, for a review).

While the conventional pathway for SMBH formation involves the collapse of massive stars, episodic periods of super-Eddington accretion (e.g., Madau et al. 2014; Jiang et al. 2019; Massonneau et al. 2023), as well as exotic theories involving primordial black holes, inflation, and dark matter have been proposed (e.g., Clesse & García-Bellido 2015; Chen et al. 2022; Davoudiasl et al. 2022; Hooper et al. 2024). Evidence from element abundance ratios and radiation from metal-poor galaxies indicates the existence of supermassive stars in the early universe (Kojima et al. 2021), which could have collapsed into IMBHs (Ohkubo et al. 2006). Various mechanisms, such as direct collapse (e.g., Loeb & Rasio 1994; Bromm & Loeb 2003; Begelman et al. 2006; Lodato & Natarajan 2006; Hosokawa et al. 2013; Ferrara et al. 2014; Umeda et al. 2016; Haemmerlé et al. 2018a, 2018b; Latif et al. 2020; Moriya et al. 2021), collisional mergers in dense clusters (Portegies Zwart & McMillan 2002; Davies et al. 2011; Lupi et al. 2014; Das et al. 2021; Rose et al. 2022; Vergara et al. 2023) or successive black hole mergers (Fragione et al. 2022a, 2022b), and dynamical interactions in young stellar clusters (Di Carlo et al. 2021; Torniamenti et al. 2022), are proposed for the formation of IMBH seeds.

Observational efforts, including the James Webb Space Telescope (JWST; Gardner et al. 2006), aim to uncover evidence supporting or refuting these black hole seeding models. The detection and study of IMBHs will play a crucial role in distinguishing between different evolutionary pathways and understanding the origins of the first SMBH seeds (Greene 2012; Ricarte & Natarajan 2018; Sassano et al. 2021; Regan et al. 2023). Advances in observational technologies promise to shed more light on this enigmatic aspect of astrophysics in the coming decades.

5.4.2. Gravitational Radiation and Lenses

Recent advancements in gravitational-wave astronomy, particularly through instruments like the Laser Interferometer Gravitational-wave Observatory (LIGO; Abramovici et al. 1992; Abbott et al. 2009), have opened new avenues for detecting IMBHs. LIGO's advanced configuration (Harry & LIGO Scientific Collaboration 2010; LIGO Scientific Collaboration et al. 2015) allows it to potentially observe the formation of IMBHs from runaway merger processes in dense NSCs (Fragione et al. 2020a, 2020b; Fragione & Silk 2020; Escala 2021) or mergers involving black holes from Population III stars (Fryer et al. 2001; Haiman & Loeb 2001; Madau & Rees 2001; Heger et al. 2003; Volonteri et al. 2003; Abadie et al. 2012; Abbott et al. 2017; Kovetz et al. 2018; Antonini et al. 2019; Wang et al. 2022). While earlier observing runs yielded no IMBH merger candidates (Abbott et al. 2017, 2019; Vajpeyi et al. 2022), recent years have seen the detection of potential IMBHs, with masses around$156.3^{+36.8}_{-22.4}\,M_\odot$ (GW190521; Abbott et al. 2021) and possibly $172.9^{+37.7}_{-33.6}\,M_\odot$ (Abbott et al. 2024, see Abbott et al. 2022). These detections, such as the gravitational-wave event GW190521, prompt speculation about their origins, including mergers within ultradwarf galaxies (Palmese & Conselice 2021) or intermediate-mass ratio inspirals involving IMBHs (Fishbach & Holz 2020; Nitz & Capano 2021).

Future ground-based observatories like the Cosmic Explorer (Evans et al. 2021) and space-based missions like the Laser Interferometer Space Antenna (LISA; Amaro-Seoane et al. 2017, 2023) hold promise for further exploring the population of IMBHs across different masses and redshifts. Particularly, modeling by Izquierdo-Villalba et al. (2023) shows that the LISA will detect coalescing massive black hole binaries with total masses of 10^4–$10^5\,M_\odot$ in host galaxies that are predominantly "extreme late-types" (i.e., akin to our Sd galaxies) at $z = 1.5$–3.0.[39] Additionally, collaborations between observatories, such as joint observations between LISA and TianQin, will enhance our understanding of these enigmatic objects (Torres-Orjuela et al. 2023). Furthermore, combining gravitational-wave observations with electromagnetic data is crucial for a comprehensive understanding of IMBHs, including their distribution and potential formation mechanisms (Saini et al. 2022; Piro et al. 2023; Bardati et al. 2024). Microlensing (Sahu et al. 2022a; Lam et al. 2022) and studies of globular clusters (Kains et al. 2016) offer complementary methods for detecting and studying IMBHs, shedding light on their prevalence and dispersal throughout the Universe.

Emerging evidence indicates that the true population is significant and that the Universe might be rife with IMBHs. Recently, Paynter et al. (2021) presented a novel way to search for and find IMBHs as the intervening gravitational lenses to distant gamma-ray bursts (GRBs). They claim to have discovered an IMBH at $z \approx 1$, which is lensing a background GRB at $z \approx 2$. Depending on the unknown redshift, they estimate the mass of the lensing object to be $\approx 5.5 \times 10^4\,M_\odot$, which they interpret as evidence for an IMBH. Based upon the observed frequency of lensed GRBs, they estimate that the present-day number density of IMBHs with masses between $\approx 10^4$ and $10^5\,M_\odot$ is $\approx 2.3 \times 10^3\,\mathrm{Mpc}^{-3}$. Compared with our aforementioned estimate of $>4.96 \times 10^{-6}\,\mathrm{Mpc}^{-3}$ for central IMBHs, this implies that the *vast* majority of IMBHs are not in the center of galaxies. Overall, these multifaceted approaches contribute to unraveling the mysteries surrounding IMBHs and their role in shaping the cosmos.

5.4.3. Star Clusters

Various astronomical phenomena, including young, globular, and NSCs, contribute significantly to black hole formation (Mapelli et al. 2021a, 2021b). NSCs play crucial roles in the hierarchical mergers of black holes, resulting in distinct merger rates and masses (see Neumayer et al. 2020, for a review of NSCs). Young dense massive star clusters rapidly produce IMBHs within 15 Myr (Rizzuto et al. 2021), covering a wide mass range observed in gravitational-wave detections (Rizzuto et al. 2022). Gas-rich NSCs act as long-lived channels continuously forming IMBHs throughout cosmic time (Natarajan 2021).

Simulations show that nucleated dwarf galaxies are prime environments for efficient IMBH mergers (Khan & Holley-Bockelmann 2021). Tidal stripping of nucleated dwarf galaxies yields ultracompact dwarf galaxies (e.g., Ferrarese et al. 2016; Graham 2020; Dumont et al. 2022; Pechetti et al. 2022), which

[39] For this endeavor, the Roman Space Telescope will be able to aid in finding precursors, i.e., massive black hole binaries before their separations are quite small enough to be detected via gravitational waves with the LISA (Haiman et al. 2023).

can ultimately become the captured nuclei of previously bulgeless galaxies via mergers, with their IMBHs in tow (Graham et al. 2021a). X-ray activity in the centers of low-mass galaxies, like those in our sample, can indicate the presence of nuclear IMBHs (Graham et al. 2019, 2021b).

Most galactic nuclei harbor SMBHs, with a significant fraction coexisting with NSCs (e.g., Sruthi & Ravikumar 2023). However, not all nucleated galaxies possess SMBHs. Askar et al. (2021, 2022) explore factors influencing the occupation fraction of SMBHs in NSCs. Gravitational-wave recoil kicks can affect SMBH presence in NSCs, especially in lower-mass galaxies (Amaro-Seoane & Freitag 2006; Gürkan et al. 2006; Amaro-Seoane et al. 2007; Arca Sedda & Mastrobuono-Battisti 2019; Arca Sedda et al. 2023). However, more massive NSCs (with escape velocities $\gtrsim 400\,\mathrm{km\,s^{-1}}$) inevitably form high-mass IMBHs (Chattopadhyay et al. 2023). SMBH formation in spheroids depends on the spheroid's mass, with only those above a certain threshold hosting SMBHs (Kroupa et al. 2020). Ultracompact dwarf galaxies may not always possess IMBHs but rather compact subclusters of normal black holes and neutron stars (Mahani et al. 2021).

5.4.4. TDEs

Observations of SMBHs are often biased toward detecting the most massive, whereas TDEs only result from $\lesssim 10^8\,M_\odot$ (see $\gtrsim 10^7\,M_\odot$, as pointed out by Coughlin & Nixon (2022); Nicholl et al. (2022), owing to a population of tidally destroyed stars that is dominated by low-mass stars) black holes (Hills 1975; Rees 1988; MacLeod et al. 2012). TDEs with IMBHs involve successive close encounters with stars, revealing the IMBH's mass (Kıroğlu et al. 2023). TDEs involving white dwarfs and IMBHs (e.g., Luminet & Pichon 1989; Rosswog et al. 2009; Clausen & Eracleous 2011; Haas et al. 2012; Cheng & Bogdanović 2014; MacLeod et al. 2016; Tanikawa et al. 2017; Vick et al. 2017; Tanikawa 2018; Maguire et al. 2020; Lam et al. 2022; Chen et al. 2023b) may indicate masses $\lesssim 10^5$, $M_\odot$ (Kobayashi et al. 2004; Rosswog et al. 2009; Gezari 2021), aiding in lower-mass black hole identification. This presents an opportunity to exclusively identify lower-mass black holes and extend black hole mass scaling relations down to lower masses (e.g., Ramsden et al. 2022). Additionally, gravitational-wave detectors can potentially observe tidal stripping events within the Local Supercluster (Chen et al. 2023b).

Black hole growth in galactic nuclei can be explained by gas and tidal disruption accretion in dense NSCs (Lee et al. 2023). Runaway tidal encounters in NSCs can lead to the formation of SMBHs (Stone et al. 2017; Baldassare et al. 2022). TDE rates aid in understanding black hole seeding mechanisms and the BHMF (Stone & Metzger 2016; Coughlin & Nicholl 2023; Yao et al. 2023). Recent observations of TDEs involving IMBHs (Perley et al. 2019; Lin et al. 2020; Wen et al. 2021; Angus et al. 2022) contribute to understanding their frequency, with growing samples in dwarf galaxies (Molina et al. 2021). TDE rates are higher in nucleated galaxies (Pfister et al. 2020), but distinguishing between complete and partial TDEs is crucial (Bortolas et al. 2023). Infrared surveys complement optical and X-ray surveys for detecting dust-obscured TDEs in star-forming galaxies (Panagiotou et al. 2023).

5.4.5. Dwarf Galaxies and Evolution

Studies examining black holes in dwarf galaxies reveal their crucial role in hierarchical galaxy evolution, indicating a shift from the traditional belief that massive black holes are solely located in giant galaxy nuclei (see Reines 2022, for a review). Occupancy of IMBHs in dwarf galaxies exceeds 50% (Greene et al. 2020), challenging previous notions. While observational searches for IMBHs in dwarf galaxies are increasing (e.g., Dong et al. 2007; Farrell et al. 2009; Reines et al. 2013; Lin et al. 2016; Ferré-Mateu et al. 2021; Mićić et al. 2022; Salehirad et al. 2022; Hatano et al. 2023, 2024), simulations struggle to replicate these findings (Haidar et al. 2022), highlighting existing discrepancies. Additionally, feedback from central black holes impacts galaxy evolution, yet inconsistencies persist between simulations (Lanfranchi et al. 2021) and observations (Davis et al. 2022).

Dwarf galaxies hosting low-luminosity AGNs suggest the presence of accreting IMBHs (Mezcua et al. 2016, 2018; Mezcua & Domínguez Sánchez 2020; Schutte & Reines 2022; Yang et al. 2023). Understanding black holes in dwarf galaxies is vital for testing formation models and predicting black hole masses (Volonteri et al. 2008; van Wassenhove et al. 2010; Silk 2017; Barai & de Gouveia Dal Pino 2019; Zaw et al. 2020; Goradzhanov et al. 2022), especially as evidence suggests dwarf galaxies' black holes may be over-massive compared to larger galaxies (Reines et al. 2011; Secrest et al. 2017; Weller et al. 2023a; Maiolino et al. 2023; Mezcua et al. 2023; Pacucci et al. 2023; Sanchez et al. 2023; Spinoso et al. 2023; Stone et al. 2023, see Pacucci et al. 2018). The IMBH occupation fraction is crucial for seeding theories (Chadayammuri et al. 2023; Spinoso et al. 2023) and extending black hole mass scaling relations to a lower regime (e.g., Graham & Scott 2013; Savorgnan et al. 2013; Savorgnan 2016a, 2016b; Savorgnan et al. 2016; Davis et al. 2017, 2018, 2019a; Sahu et al. 2019a; Davis et al. 2019b; Sahu et al. 2019b; Davis et al. 2019c; Sahu et al. 2020; Davis et al. 2021; Sahu 2021; Sahu et al. 2022b, 2022c; Sahu 2022; Graham & Sahu 2023a, 2023b; Graham 2023b, 2023c; Davis & Jin 2023; Jin & Davis 2023). Improved understanding of black hole–galaxy coevolution aids in calculating the black hole mass density in the Universe (e.g., Graham et al. 2007; Davis et al. 2014; Mutlu-Pakdil et al. 2016), elucidating their evolutionary history, and ascertaining causal mechanisms (Pasquato et al. 2023).

5.5. Future Work and Possible Observations

In their investigation of the accretion disks around IMBHs, Ogata et al. (2021) predicted that IMBHs could be observed as bright sources in the infrared band. Furthermore, Cann et al. (2018, 2021) have shown that infrared coronal lines can find AGNs even when optical diagnostics fail, as demonstrated in their study of an optically normal, low-metallicity dwarf galaxy. As such, coronal lines are ideal and the infrared JWST will be well suited for spotting IMBHs (Reefe et al. 2022, 2023, see Herenz et al. 2023). Armed with superior sensitivity, resolution, and spectroscopic multiplexing capabilities, the ongoing JWST observations will provide a boon to scientific discovery (Kalirai 2018).

Impressively, the JWST has quickly found confirmed galaxies up to $z = 13.2$ (Curtis-Lake et al. 2023) and candidate objects up to an incredible $z \approx 20$ (Yan et al. 2023), as well as surprisingly massive galaxies (up to $\sim 10^{11}\,M_\odot$) at $7.4 < z < 9.1$ (Labbé et al. 2023). Qin et al. (2023) found that boosted star formation

efficiencies and reduced feedback regulation are necessary to reproduce $z \gtrsim 16$ JWST galaxy candidates, which are susceptible to low-redshift contamination from $z \sim 5$ galaxies. See also the cautionary report from Zavala et al. (2023) that detailed how dusty starbursts at lower redshifts can masquerade as ultrahigh photometric redshift galaxies in JWST observations. Furthermore, it is worth investigating the implications of bursty star formation at high redshifts leading to selection effects and associated biases for the JWST (Looser et al. 2023; Sun et al. 2023). Already testing the onset of spiral galaxy formation, the JWST has helped identify a dusty multiarm spiral galaxy at $z = 3.06$ (Wu et al. 2023) and found that disk galaxies are prevalent out to at least $z = 9.5$ (Kartaltepe et al. 2023; Sun et al. 2024) Similarly, the JWST has enabled the identification of the most distant barred galaxies to-date, out to $z \simeq 3$ (Costantin et al. 2023; Guo et al. 2023; Huang et al. 2023). Of particular interest to black hole seeding and early BH–galaxy assembly, the JWST has discovered a crop of low-mass (undermassive; Stone et al. 2024) galaxies at high redshifts (Williams et al. 2023a; Übler et al. 2023; Curtis-Lake et al. 2023; Gelli et al. 2023; Kocevski et al. 2023; Robertson et al. 2023; Strait et al. 2023; Looser et al. 2024).

The JWST has the ability to color-select black hole seeds transitioning to SMBHs (Goulding & Greene 2022) and distinguish rest-frame optical lines for the identification of "light seed" Population III (Vanzella et al. 2023) and "heavy seed" direct-collapse black holes in the early Universe (Nakajima & Maiolino 2022). Models estimate that the JWST surveys will have the sensitivity to detect heavy black hole seeds out to redshifts of $z \lesssim 14$ (Trinca et al. 2023). Already, the JWST has spotted high-redshift AGNs (Larson et al. 2023; Maiolino et al. 2023, 2024) and established constraints on their formation that require either super-Eddington accretion from a stellar-mass seed or Eddington accretion from a very massive black hole seed. Moreover, the JWST has already witnessed an X-ray quasar ($M_\bullet \sim 4 \times 10^7\ M_\odot$ in a comparably massive host galaxy) at $z = 10.3$, which "suggests that early SMBHs originate from heavy seeds" (Bogdán et al. 2023; Goulding et al. 2023; Natarajan et al. 2024) and constitutes the first outsize (née obese) black hole galaxy (Natarajan et al. 2017). Radio observations from the forthcoming Square Kilometer Array could also detect emissions from direct-collapse black holes at high-z (Whalen et al. 2020, 2023). The parameters for black hole growth and seed models will become significantly constrained as future observations discern $z > 8$ quasars (Pacucci & Loeb 2022).

In order to dynamically confirm the IMBH mass estimates we have made, it would be necessary to resolve kinematics within the gravitational sphere of influence (SOI) of our black holes. From Peebles (1972),[40] a black hole at the center of a galaxy has an SOI with a radius $r_h \equiv GM_\bullet \sigma_0^{-2}$. From our sample of 23 targets of interest, UGC 3949 is the only galaxy with a known central velocity dispersion ($\sigma_0 = 70.1 \pm 4.9$ km s^{-1}). For UGC 3949 ($d = 89.9 \pm 6.9$ Mpc and $M_\bullet = 5.05 \pm 2.99 \times 10^4$ M$_\odot$), we obtain $r_h = 44.2 \pm 26.8$ mpc $= 101 \pm 62$ μas.

With its stunning resolution of 20 μas, the Event Horizon Telescope (EHT) could resolve this region in the center of UGC 3949. The EHT resolved the emission rings surrounding the SMBHs M87* and Sgr A* with diameters of $\approx$42 μas (Event Horizon Telescope Collaboration et al. 2019; Medeiros et al. 2023) and $\approx$52 μas (Event Horizon Telescope Collaboration et al. 2022), respectively. Indeed, distance alone would not be a hindrance in resolving similarly sized IMBHs in our sample of targets, which only extends out to about 125 Mpc. Granted, even if the SOI were resolved in images of our targets, spectra would still be useful.

We hope to target the 23 candidates with follow-up spectroscopy (for stellar velocity dispersions, gas emission lines, and black hole virial mass estimates),[41] X-ray imaging (for AGNs activity), and perhaps simultaneous radio interferometry (for the fundamental plane of black hole activity; Merloni et al. 2003; Falcke et al. 2004). However, Gültekin et al. (2022) cautioned against the use of the fundamental plane of black hole activity for identifying IMBHs without additional constraints beyond just straightforward X-ray and radio observations. Furthermore, only a small fraction ($\sim$0.6%) of IMBH-candidate host galaxies are radio-band active (Yang & Yang 2023). Further X-ray information such as the characteristic variability or the normalized excess variance (variability amplitude) can also be used to estimate black hole masses of AGNs (Pan et al. 2015). However, the characteristic damping timescale at X-ray wavelengths is substantially shorter (González-Martín & Vaughan 2012), and less correlated with black hole mass, than at optical wavelengths (Burke et al. 2021).[42] Additionally, the variability timescale at submillimeter wavelengths appears to also be a useful parameter that correlates well with black hole mass in low-luminosity AGNs (Chen et al. 2023a).

6. Conclusions

The comparisons throughout this work seem to indicate that Sd galaxies exhibit characteristics that are surprisingly similar to the general population of spiral galaxies, in the aggregate, although we did not explore bulge mass, as many Sd galaxies are (considered) bulgeless. However, we note that a range of B/T flux ratios exist at each morphological type (Graham & Worley 2008). This unexpected resemblance could be a symptom of subjective morphological misclassifications, or perhaps is showing a natural diversity even among this subpopulation of apparent late-type spiral galaxies. Specifically, we find the expectation that an archetypical Sd should have a low mass, be slowly rotating, display loosely wound spiral arms, and have a low-mass central hole is not unimpeachable; characteristics more akin to earlier-type spiral galaxies appear to be endemic. Thus, the classical Sd morphological class is a stereotype; there is no average Sd galaxy. In any regard, we do find merit in exploring Sd galaxies for IMBHs, as they possess the requisite environmental traits (for hosting IMBHs) in a higher proportion than the average spiral galaxy.

On the whole, we find that a randomly selected Sd galaxy will have a 27.7% chance of possessing an IMBH. Our search has produced 23 candidates, each with a probability of at least 50% of hosting an IMBH. Although we expect 23/85 of our galaxies house an IMBH, the product of all their $P(\mathcal{M}_\bullet \leqslant 5)$ probabilities implies an $\approx$100% certainty that at least one of our 85 Sd galaxies possesses an IMBH. We intend to make these targets the focus of continued research. The fruition of the

[40] See also the corresponding Section 3.3 from Davis & Graham (2021).

[41] Cho et al. (2023) elucidate how better constraints for the Hα size–luminosity relation are required to calibrate a virial mass estimator based on the Hα broad emission line from low-luminosity AGNs and IMBHs.

[42] Although Treiber et al. (2023) revealed the substantial effect of contamination from variable stars in their search for stochastic variability.

long-sought quest to identify IMBHs will finally complete the gap in our knowledge of the demography of black holes.

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Facilities: CXO, GALEX, PS1, Sloan.

Software: `2DFFT` (Davis et al. 2016), `2DFFT Utilities`, `2DFFTUtils Module`, `Astropy` (Astropy Collaboration et al. 2013, 2018), `ChatGPT`, `Hyper-Fit` (Robotham & Obreschkow 2015, 2016), `IRAF` (Tody 1986, 1993), `K-corrections calculator`, `Matplotlib` (Hunter 2007), `NumPy` (Harris et al. 2020), `Pandas` (McKinney 2010), `pearspdf`, `PySR` (Cranmer 2023), `Python` (Van Rossum & Drake 2009), `SAOImageDS9` (Joye & Mandel 2003), `SciPy` (Virtanen et al. 2020), `SpArcFiRe` (Davis & Hayes 2014), `Spirality` (Shields et al. 2015), `uncertainties`.

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